Department of Artificial Intelligence,
Sogang University

Paper seminar: Audio-Aware Robot Policies

From generative audio to streaming auditory memory

Gwangwoong Park

Business Administration &
Computer Science and Engineering

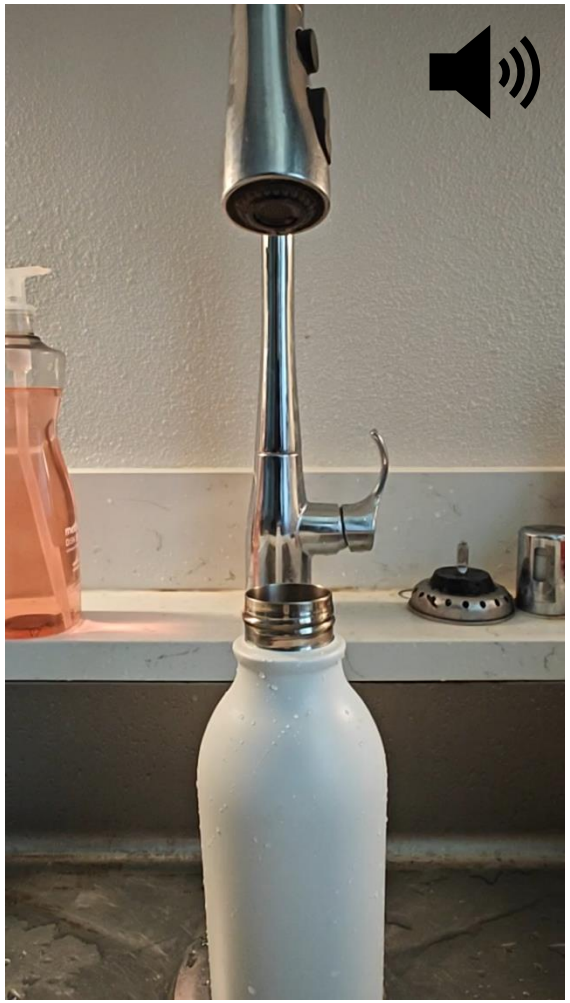
NLP & ISDS Lab

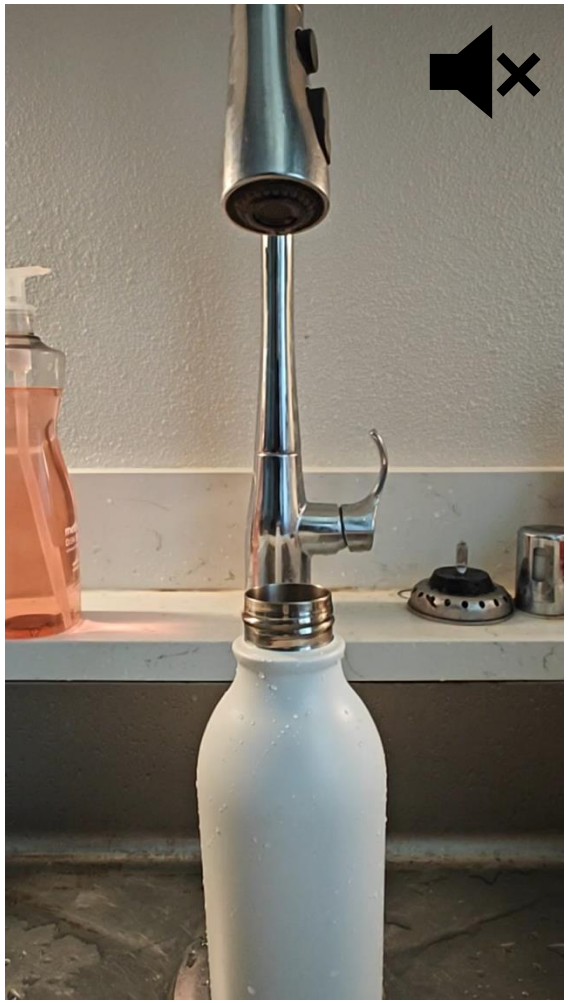
2026.11.04

Agenda

I	Background: Why sound in robot policies?
II	Paper 1: MultiGen (generative audio for sim-to-real)
III	Paper 2: HEAR (streaming auditory memory)
IV	Comparison and critical discussion
V	Open questions and summary

The “Say When” Task





Source: https://multigen-audio.github.io

Why sound?

Vision shows where; sound shows “when and what state”



Process state

Pour into opaque cup

→ **fill level**



Completion

Microwave “ding”

→ **timing**



Contact

Drop cube on plate

→ **timing, material**



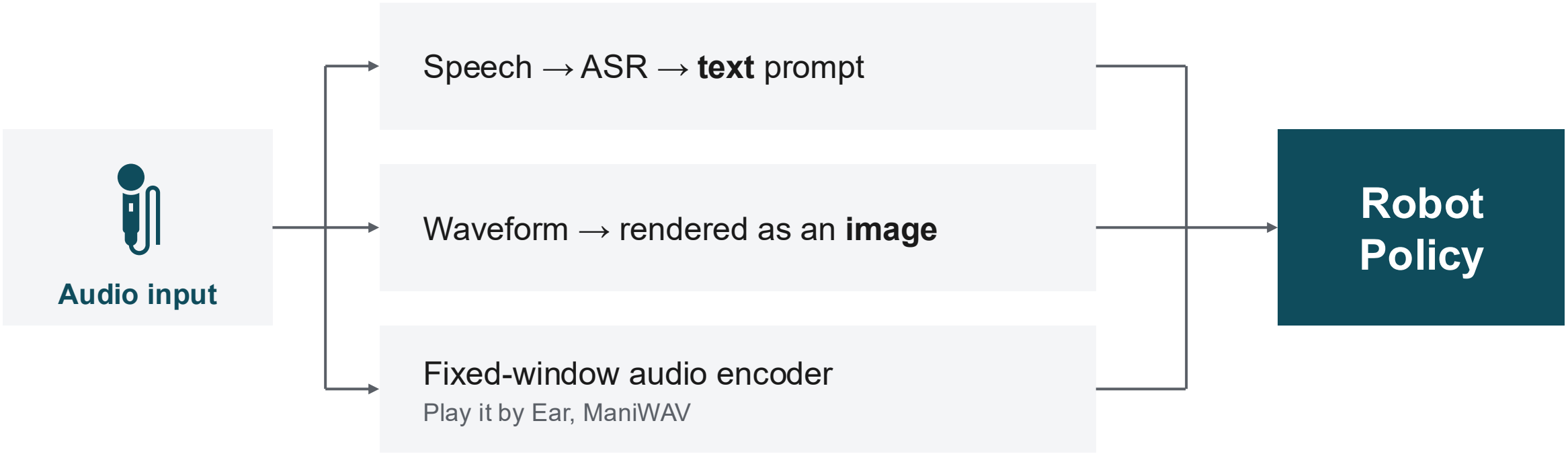
Hidden contents

Shake a bottle

→ **what and how much is inside**

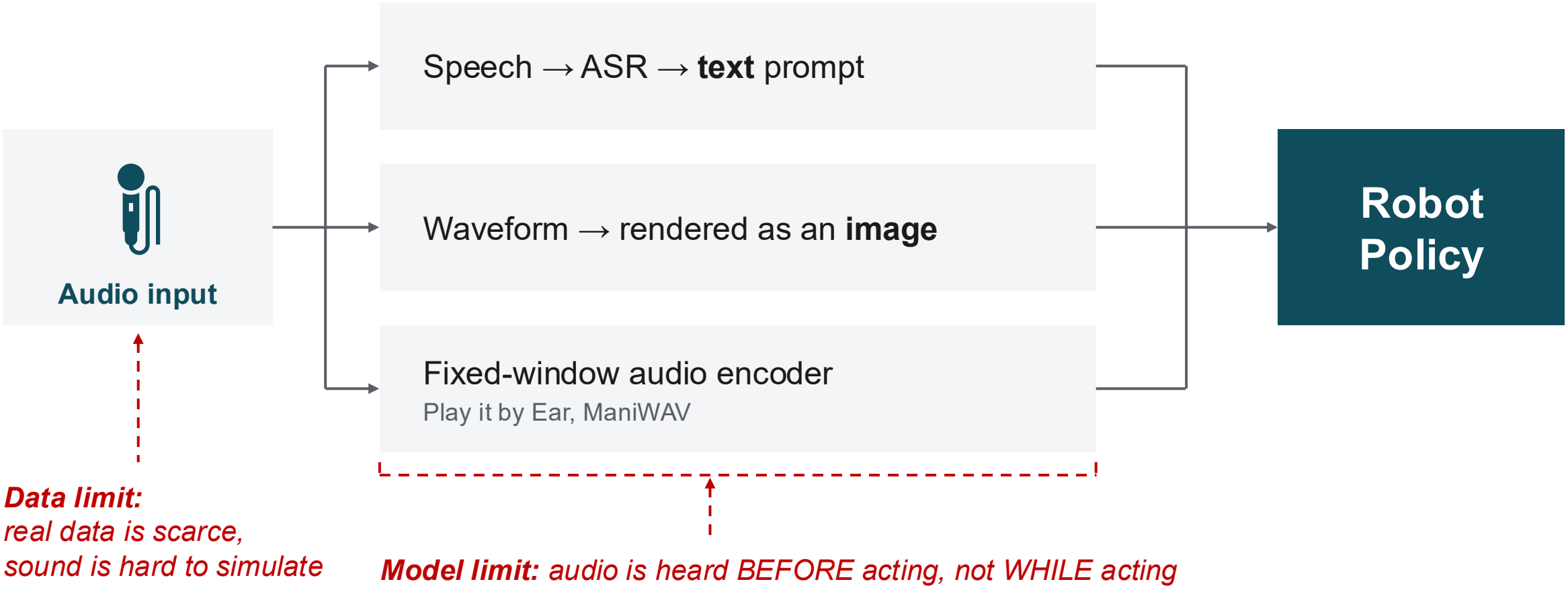
Source: HEAR, IJRR 2026

How is audio used in robot policies today?



Source: HEAR, IJRR 2026 (grouping follows HEAR); MultiGen, CoRL 2025

Two main issues..



Source: HEAR, IJRR 2026 (grouping follows HEAR); MultiGen, CoRL 2025

Two papers dealing with two issues

	Bottleneck	Paper
Data	Sound is hard to simulate	**MultiGen** (CoRL 2025) Generative audio for sim-to-real
Architecture	Short sounds are missed while acting	**HEAR** (IJRR 2026) Streaming auditory memory

Source: MultiGen, CoRL 2025 (arXiv 2507.02864); HEAR, IJRR 2026

The Sound of Simulation: Learning Multimodal Sim-to-Real Robot Policies with Generative Audio (MultiGen)

Renhao Wang et al. · UC Berkeley

CoRL 2025 · Best Paper Finalist (Top tier conference in Robotics)

arXiv 2507.02864 (Jul 2025) · multigen-audio.github.io

TL;DR

Generate audio in simulation →
train audio-visual policies with no real robot data

Source: CoRL 2025 awards page (2025.corl.org/program/awards), arXiv

Why is multimodal sim-to-real stuck?

Real data is hard

Scarce and costly

→

Use simulation!

→

Sound is hard to simulate..

Vision is near-real; simulated sound is limited

Source: MultiGen, CoRL 2025, §1, §3.1

Then.. How about using generative model?

Sound is hard to simulate →

Simulation

Physics engine

Scene, dynamics, kinematics

Generative model

Generates the sound in Sim using V2A model

→

MultiGen

Generation in Sim

Deployment in Real

Source: MultiGen, CoRL 2025, §1, §3.1

MultiGen: simulate with physics, generate what physics can't

Physics engine
Scene, dynamics, kinematics

Generative model
Generates the sound in Simulation

Zero-shot transfer
Real robot manipulation w/o real data

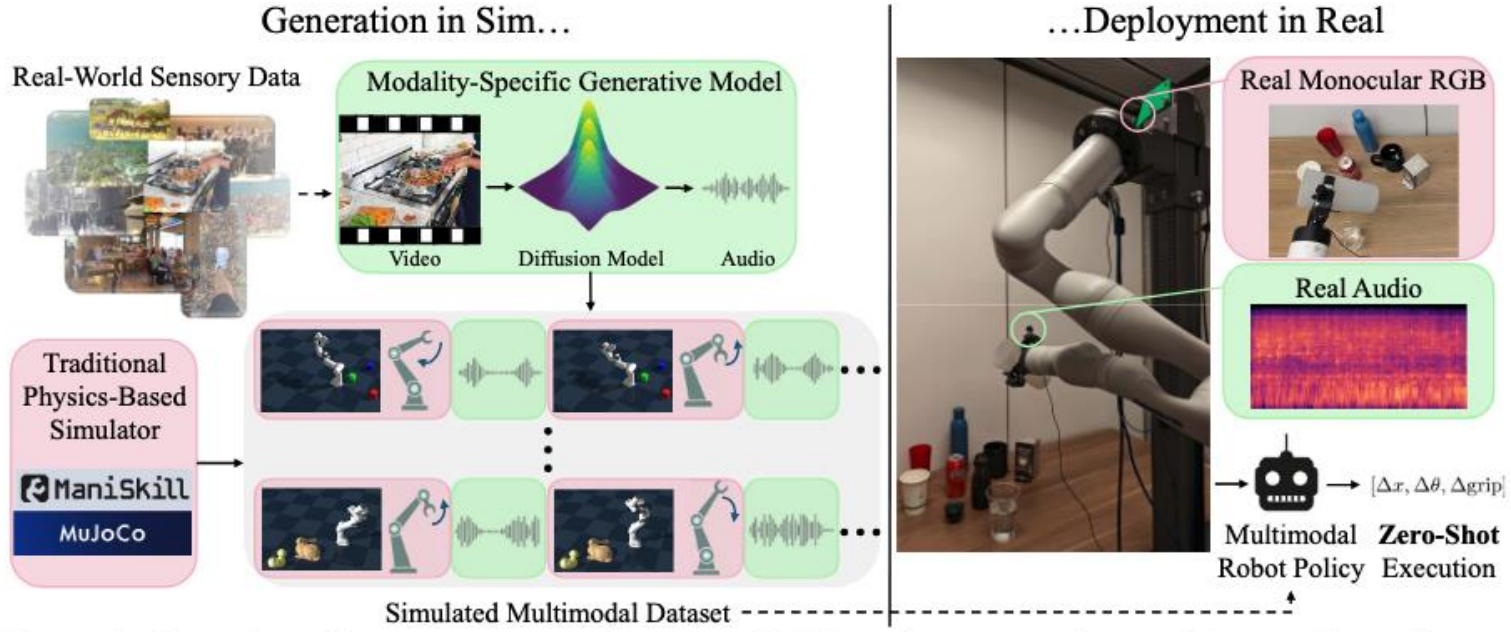


Figure 1: **Overview of our MULTIGEN framework.** We train a generative model on real-world sensory data to capture modalities that would otherwise be difficult to simulate (e.g. audio). Augmenting traditional simulators with these generative models enables generating synthetic multimodal data at scale, and learning multimodal policies that can translate *zero-shot* to the real world.

Source: MultiGen, CoRL 2025, §2–3.1

MultiGen pipeline overview

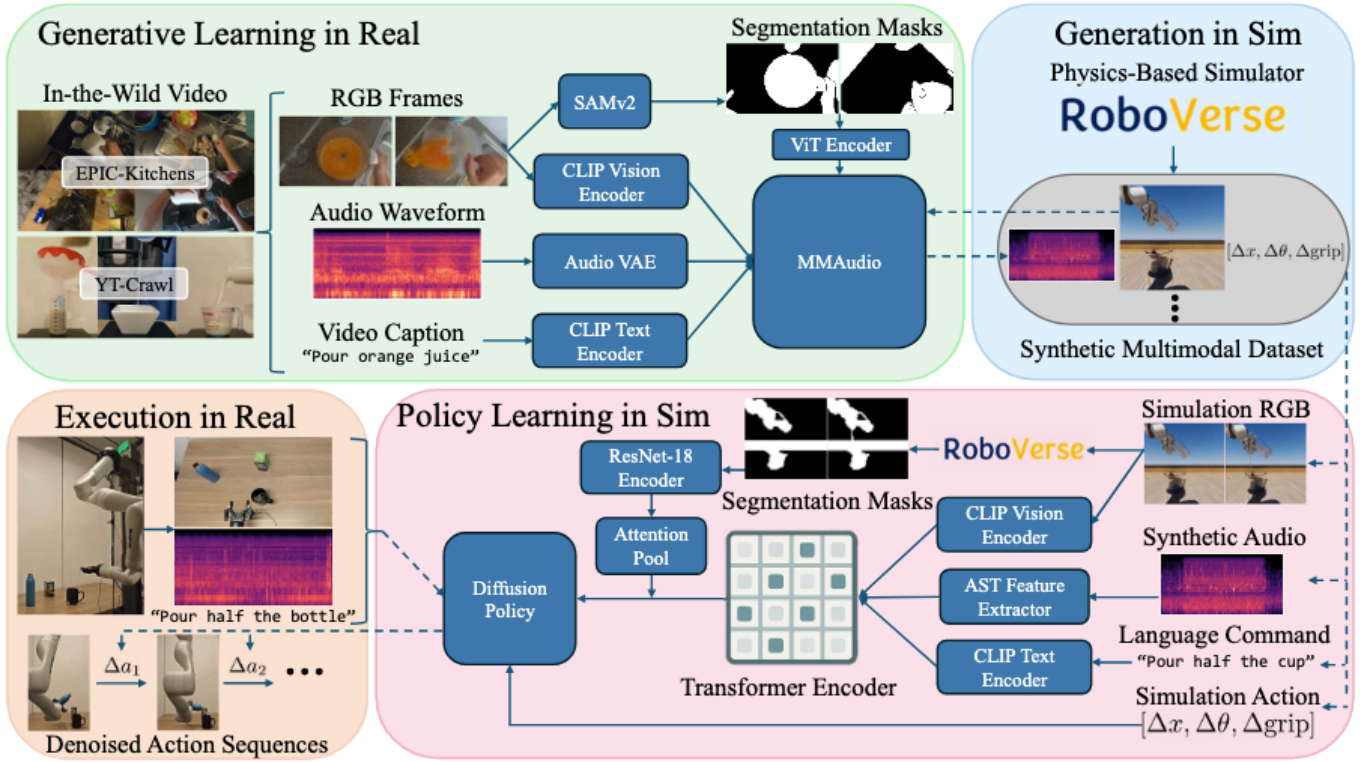


Figure 2: **Components in the MULTIGEN instantiation for robot pouring.** We first finetune a video-to-audio diffusion model (e.g.MMAUDIO) on in-the-wild video. Conditioning on simulation video then enables generating multimodal simulation trajectories replete with audio. We then train a policy (e.g. Diffusion Policy) on this multimodal dataset, before evaluating zero-shot on a real setup.

Source: MultiGen, CoRL 2025, §2–3.1

Task: liquid pouring

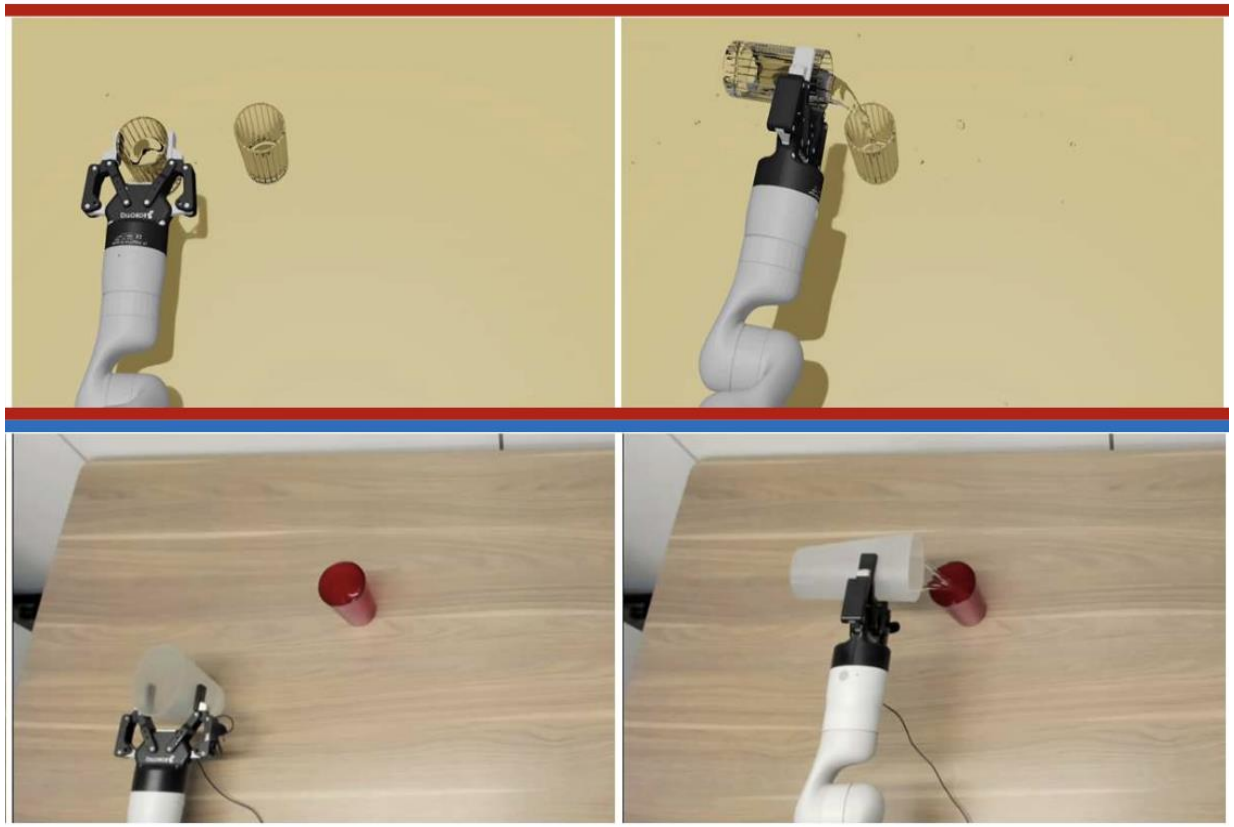
TASK

Pour a target amount into a receiving container

WHY POURING?

Occlusion, transparency, visual ambiguity

→ audio matters!



Source: MultiGen, CoRL 2025, §1, §3.3

Choice of audio generative model

	AudioLDM text-to-audio	**MMAudio** video-to-audio
Pour height	X	O
Container volume	X	O
Liquid properties	X	O

Source: MultiGen, CoRL 2025, §3.2

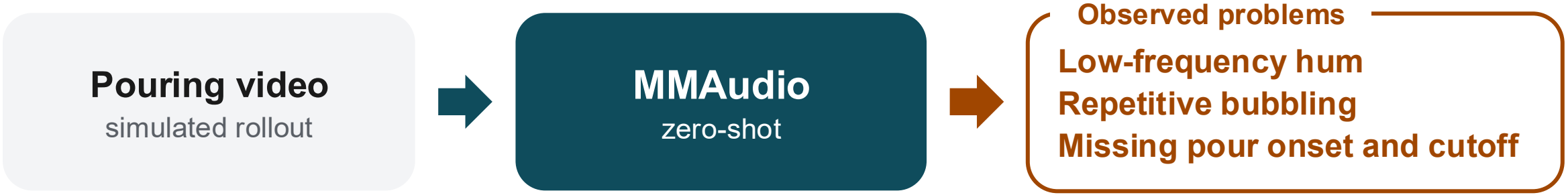
Problem 1: Pretraining data misses pouring sounds

Diverse but suboptimal data

Noisy, low-frequency Foley sounds, far from liquid sounds

Pouring is long-tail

Few instances in WavCaps, VGGSound, AudioCaps



Source: MultiGen, CoRL 2025, §3.2, Appendix B.2

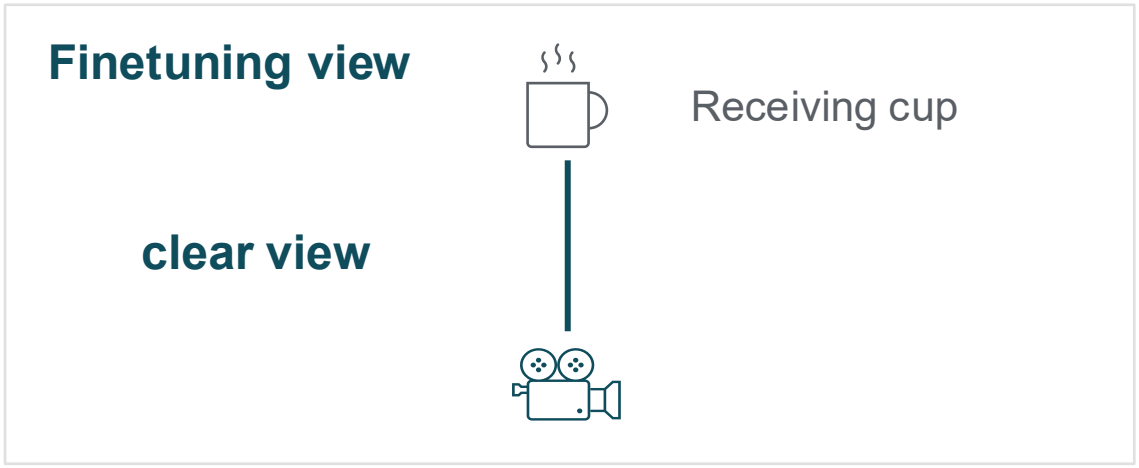
Problem 2: Simulated video looks different

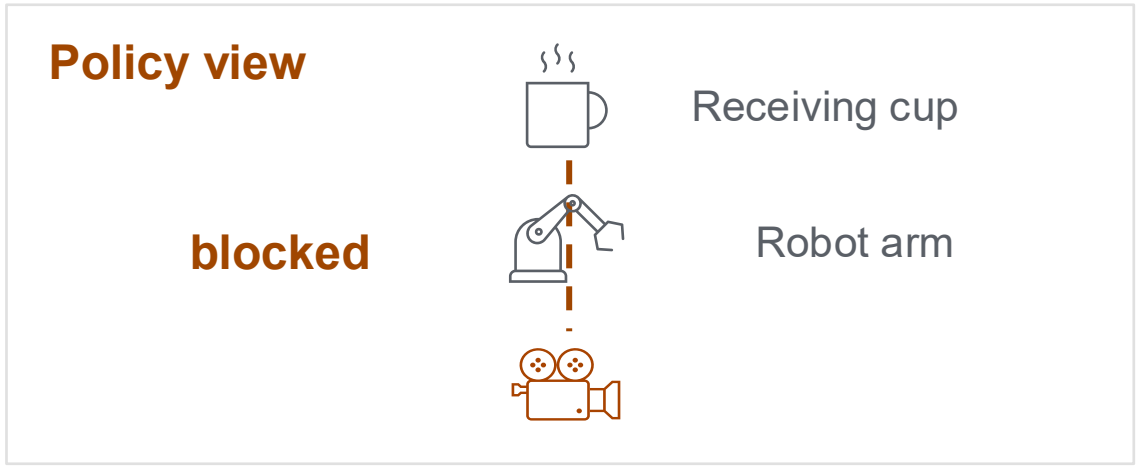
Visual distribution

Trained on natural video;
frames in simulation look unlike real

Camera views

Policy view can be occluded or misaligned with finetuning videos





Source: MultiGen, CoRL 2025, §3.2–3.3

Fix 1: Finetune on real pouring videos

Teach the generator what real pouring sounds like



EPIC-Kitchens
first-person cooking videos with many liquid manipulations



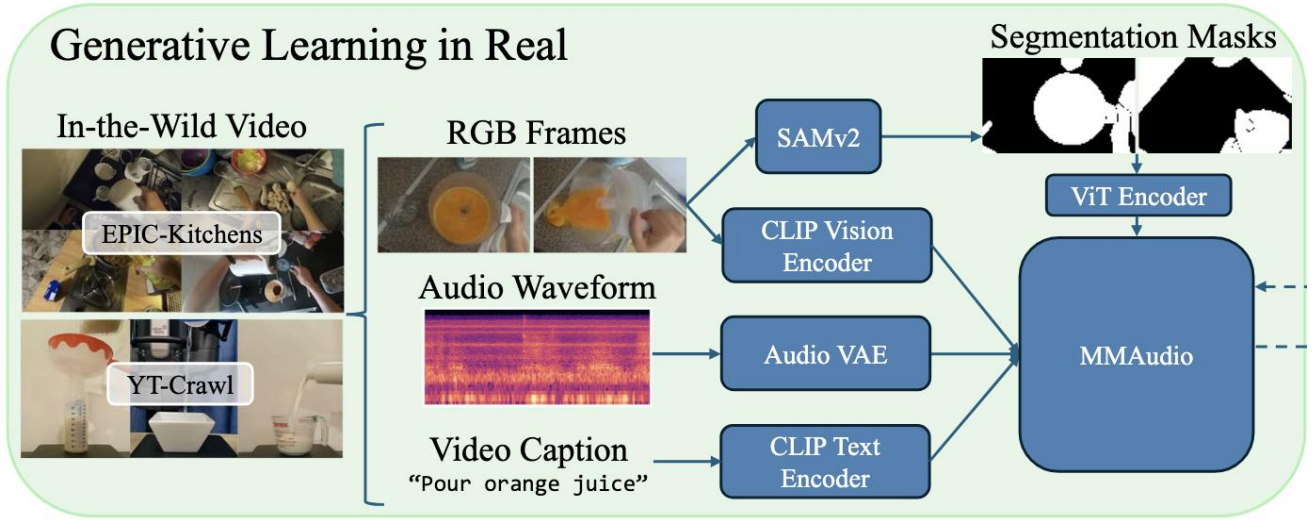
Own YouTube crawl
water, coffee, soda, soup in glass, plastic, ceramic

1,031 clips
real pouring clips, 5–60 s each

Goal: capture the nuances of real pouring sounds to generalize to new conditions

Source: MultiGen, CoRL 2025, §3.2

Fix 2: Add masks and a dedicated camera



SAMv2 masks

marks cup and liquid regions in every frame

→ **focus on meaningful regions, not the environment as a bridge between sim and real**

Gated fusion

blends segmentation mask features with CLIP vision features using a learned weight

→ **adds mask cues without losing the pretrained path**

Virtual camera

places a virtual camera in simulation to capture the receiving container

→ **matches the viewpoint of real pouring videos**

Source: MultiGen, CoRL 2025, §3.2–3.3, Appendix B.1–B.2

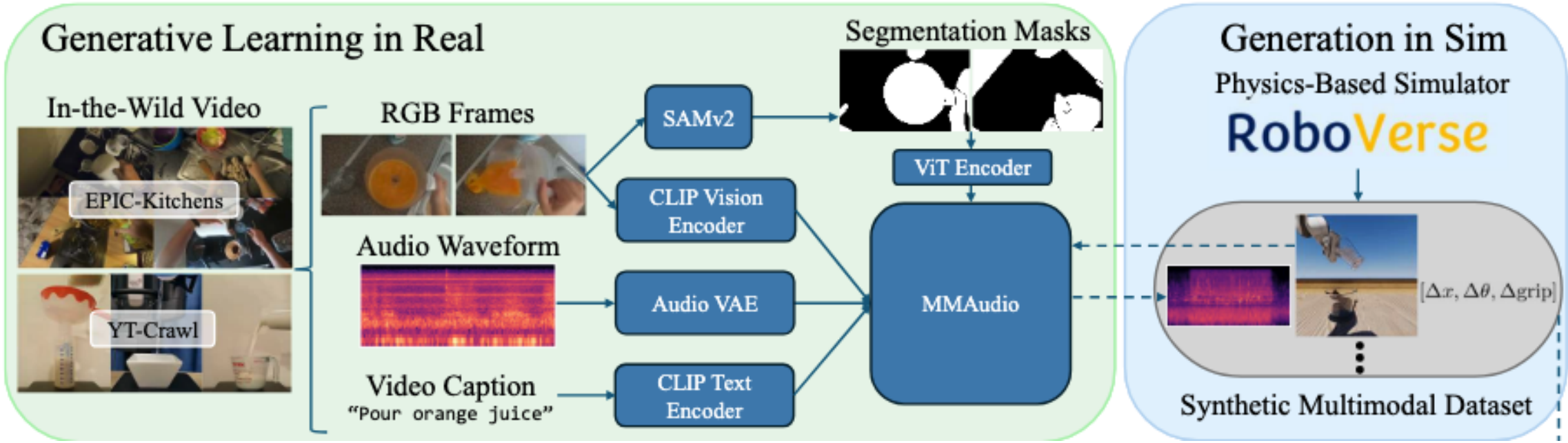
MMAudio Finetuning setup

- **Frozen:** CLIP vision encoder
- **Updated:** transformer blocks, audio decoder, segmentation module
- **Optimization:** AdamW, lr 1e-5, cosine decay, 500-step warmup, batch 32, 100K steps (~40 ep.)
- **Loss:** $L = L_{L1} + L_{SC} + L_{AudioCLIP}$ (MMAudio default terms)
- **Audio:** log-mel 64 bins, 16 kHz; DDIM sampler, 50 steps

Source: MultiGen, CoRL 2025, §3.2–3.3, Appendix B.1–B.2

Generative model adds sound to simulation

MMAudio generates matching sound from RoboVerse video

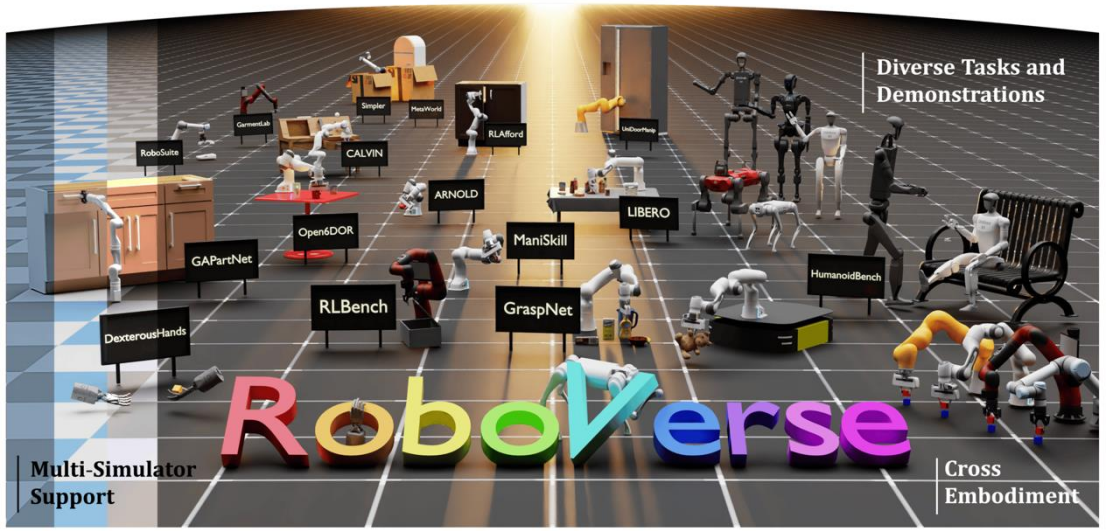


Output: frames + action + **audio (generated)**

Source: MultiGen, CoRL 2025, §3.3

Why RoboVerse

RoboVerse: a scalable, parallel simulation platform



Unified platform

Multiple simulators, one interface

Parallel execution

Large-scale trajectory generation

Assets + variety

Cups, bottles, liquids; domain randomization

Source: MultiGen, CoRL 2025, §3.3; RoboVerse, arXiv:2504.18904

Simulation environment setup

Rough setup, broad randomization: policies generalize to unseen real scenes

1. Simulation env roughly matches the real setup

Rough camera position; exact pose and lighting not needed

2. Structured domain randomization

Liquid

Density, viscosity, color, transparency

Camera pose

Small shifts in position and rotation

Lighting

Intensity, shadows, reflections

Textures

Table and background

Object scale

Cup opening, bottle height

Source: MultiGen, CoRL 2025, §3.3

Trajectory synthesis: simulated pouring

A motion planner generates the robot's pouring trajectories

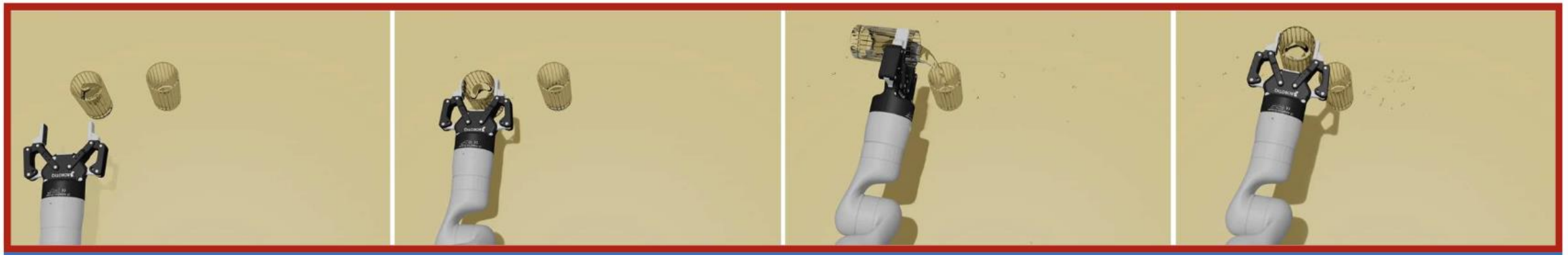
1
Pre-fill
70% ± 5% capacity

2
Grasp
Random point within 5% of center

3
Move above
Random pour height over the receiver

4
Pour
1/4, 1/2, 3/4, or full; deterministic tilt angle

5
Untilt
Episode ends

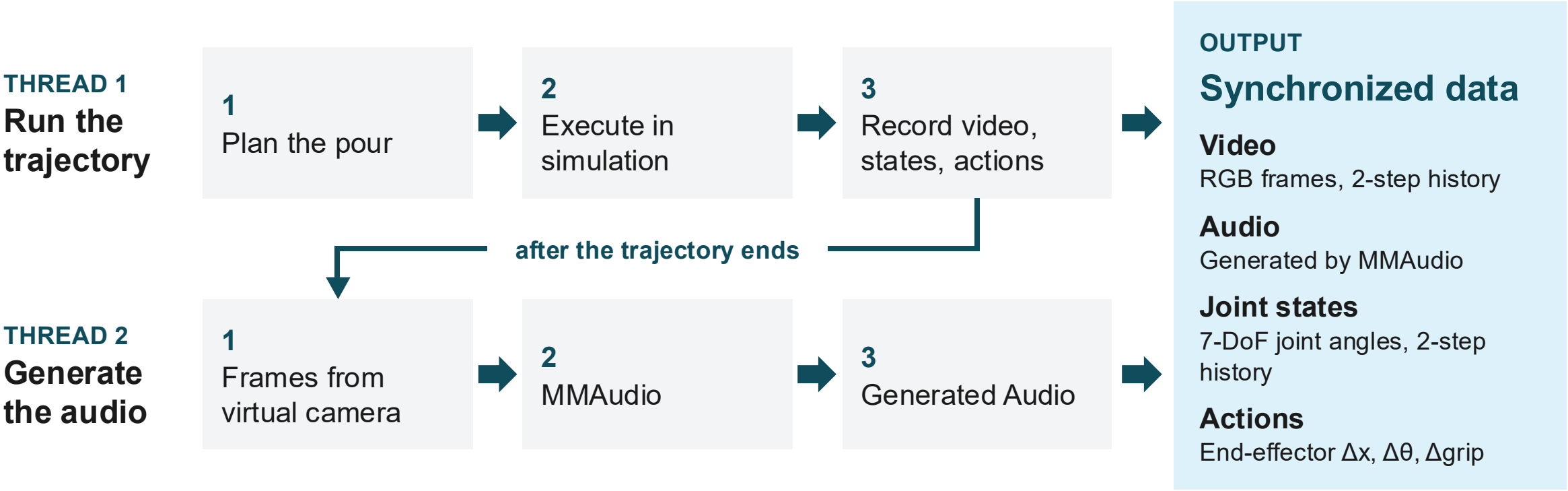


Source: MultiGen, CoRL 2025, §3.3, Fig. 3

Build synchronized sim w/ audio data

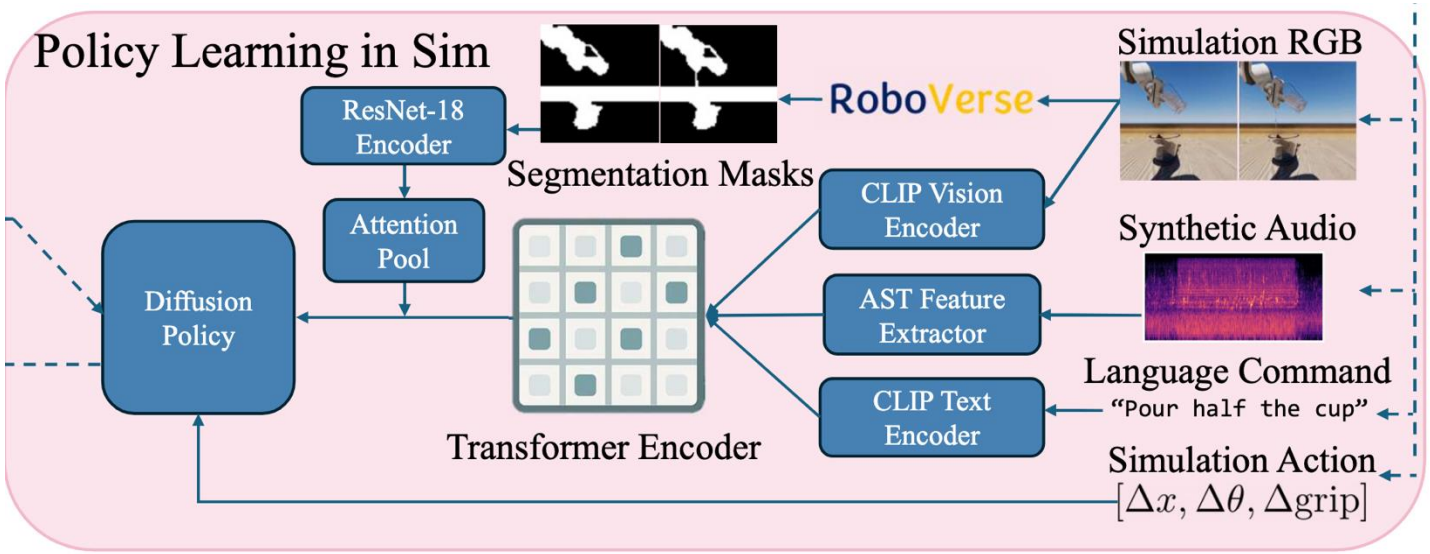
Physical engine: simulates the pour and supplies the video, joint states and actions

Generative model: MMAudio creates the matching sound, which a simulator cannot produce



Source: MultiGen, CoRL 2025, §3.3

A diffusion policy trained in simulation w/ audio



Model
- Diffusion Policy, from scratch

Input
- RGB (CLIP vision)
- audio (AST)
- Instruction (CLIP text)
- joint states (proprioception)
- Segmentation masks (attention pool)

Output
- End-effector actions
- 10 Hz, 16-step horizon
- 10-dim: 3 Δx position, Δθ 6D rotation, Δgrip

Loss
- Noise prediction (x_0: simulated actions)

Loss: standard diffusion objective, not stated in the paper. Source: MultiGen, CoRL 2025, §4.1–4.2

Zero-shot evaluation on a real robot w/ audio vs. w/o audio:

Robot
- Kinova Gen3 (7-DoF arm) + Robotiq 2F-85 (gripper)

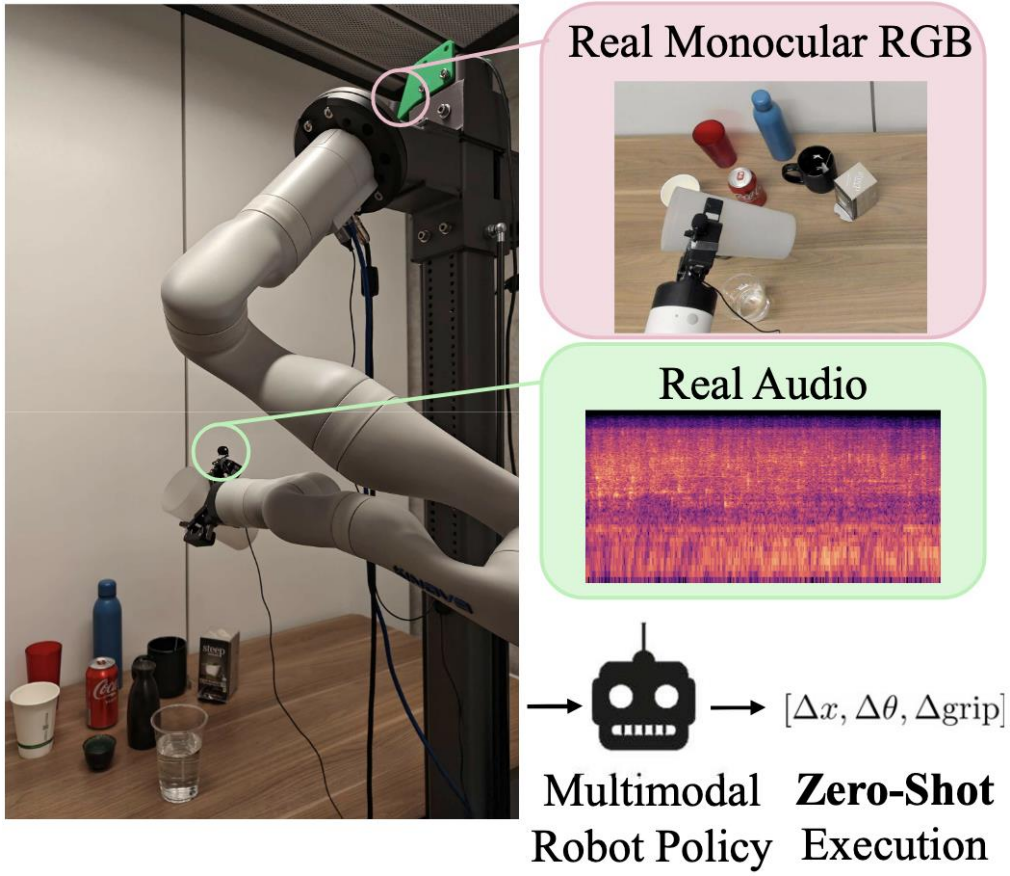
Sensors
- RGB camera
- end-effector lapel mic (192 kHz → 16 kHz)

Environment:
- Occlusion: 4 opaque, 4 transparent
- Container: plastic, paper, metal, glass
- Liquid: water, juice, soda, hot

Protocol
- 12 evaluations per condition (3 start pairs × 4 commands)

Metric

$$NMAE = \frac{|\text{Actual Poured Amount - Desired Target Amount}|}{\text{Desired Target Amount}}.$$



Source: MultiGen, CoRL 2025, §4.1–4.2, Appendix A

Main evaluation: Audio decreases pouring error

Opaque	water-white cup-red cup	coffee-metal thermos-paper cup	sake-sake carafe-sake cup	water-metal thermos-metal mug
DIFFUSION POLICY(V)	0.54 ± 0.23	0.54 ± 0.20	0.68 ± 0.30	0.47 ± 0.29
DIFFUSION POLICY(V + A)	**0.44 ± 0.19**	**0.33 ± 0.17**	**0.43 ± 0.28**	**0.37 ± 0.15**
Transparent	**water-white cup-plastic cup**	**juice-plastic bottle-plastic cup**	**soda-metal can-plastic cup**	**juice-plastic bottle-glass mug**
DIFFUSION POLICY(V)	0.53 ± 0.20	0.46 ± 0.20	0.43 ± 0.20	0.49 ± 0.21
DIFFUSION POLICY(V + A)	**0.42 ± 0.27**	**0.38 ± 0.21**	**0.39 ± 0.22**	**0.43 ± 0.18**

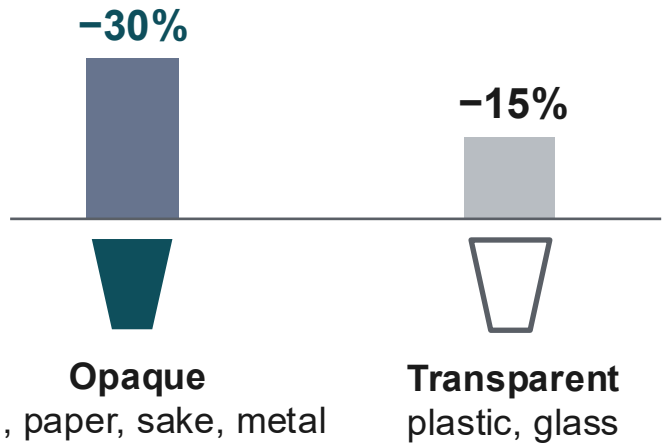
Table 1: **Main evaluation results on our pouring benchmark (lower is better).** We report average normalized mean absolute error (NMAE) and 1 std. dev. error. The top four tasks all involve opaque containers, and the bottom four tasks involve translucent containers. Results are computed over twelve seeds (four language commands evaluated for three random locations each.)

Consistently high

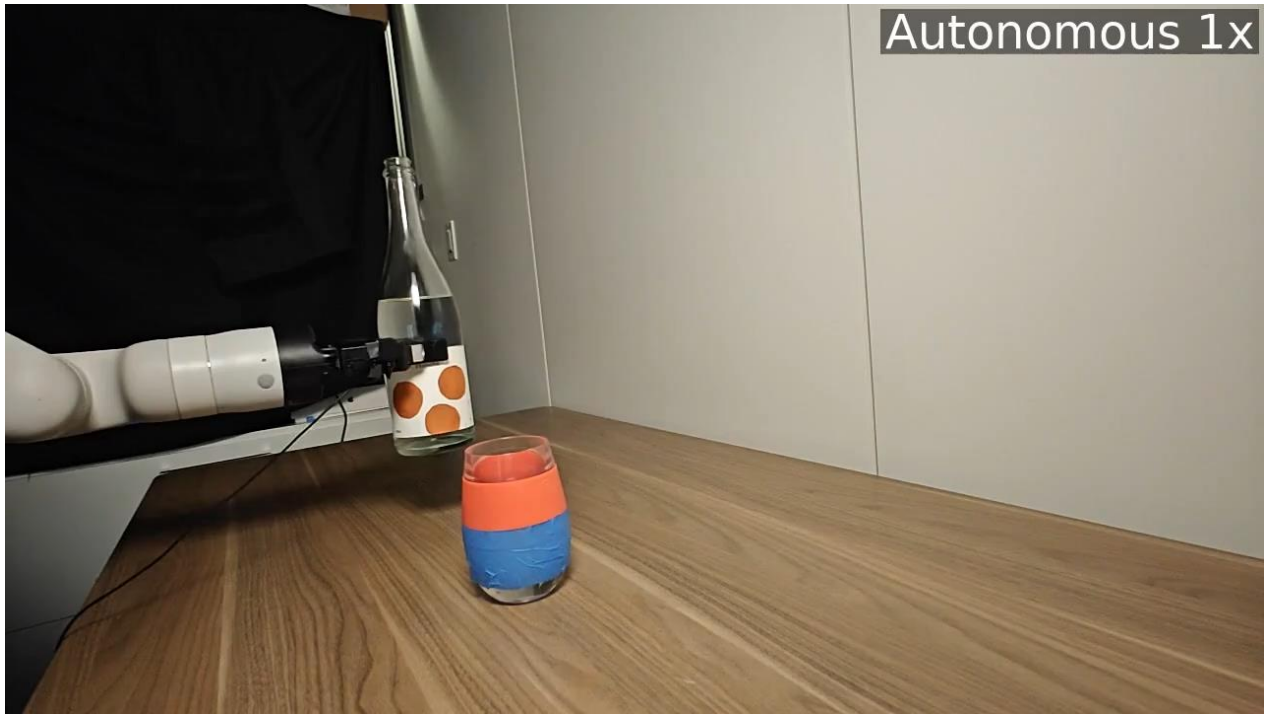
Audio beats vision-only in every condition

Opaque gains most

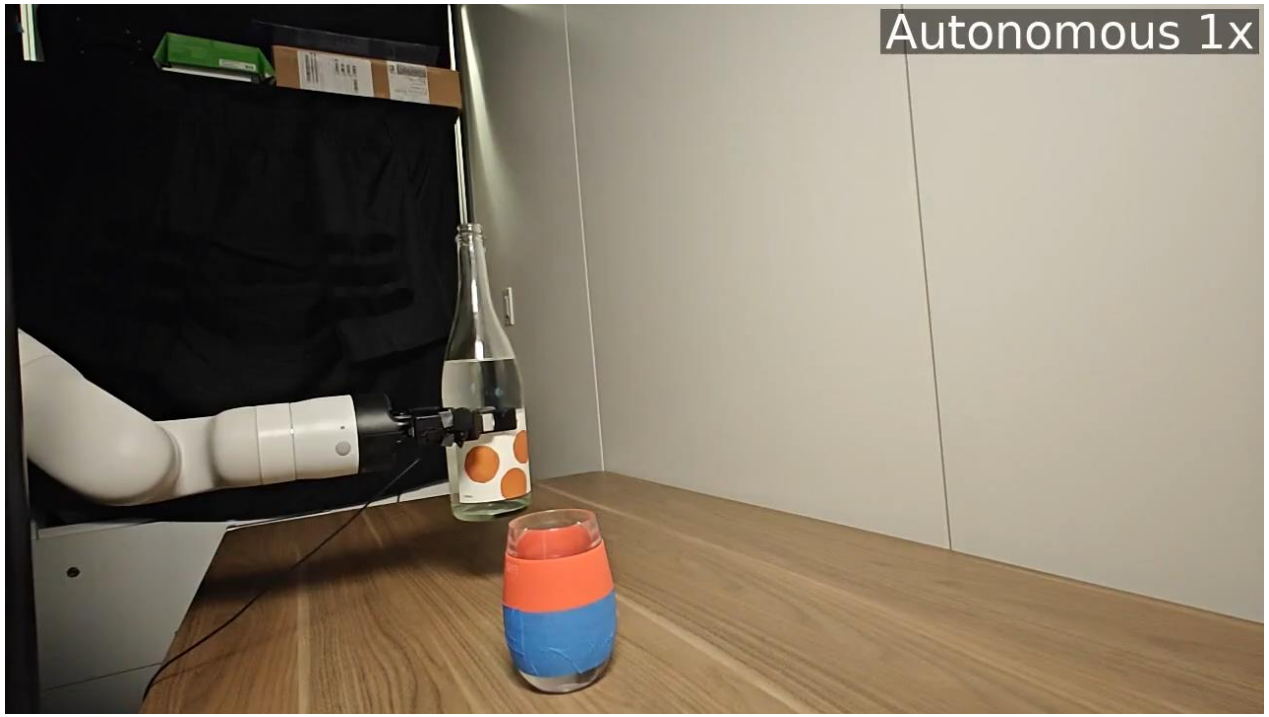
About 30% lower NMAE vs. 15% on transparent



NMAE: lower is better. Opaque/transparent = receiving container. % computed from Table 1 means. Source: MultiGen, CoRL 2025, Table 1



Vision Only



Vision + Audio

Source: https://multigen-audio.github.io

Audio quality evaluation

Results of usual robotics data augmentation method vs. a generative model

Baseline – noise augmented

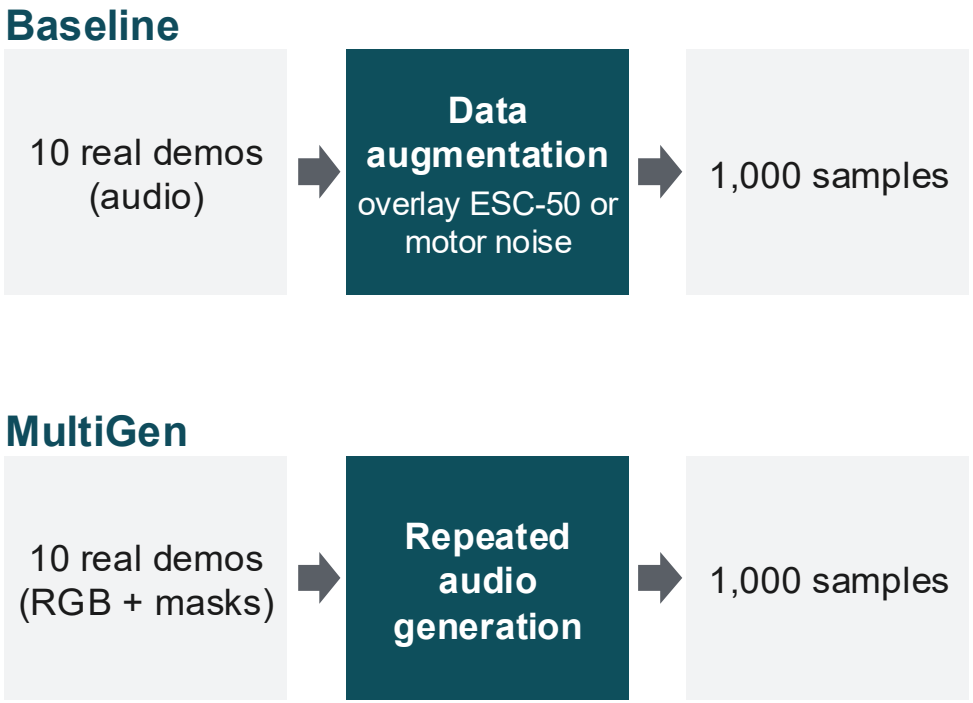
Real demo audio + data augmentation (added noise)

MultiGen

Audio generated by the finetuned generative model

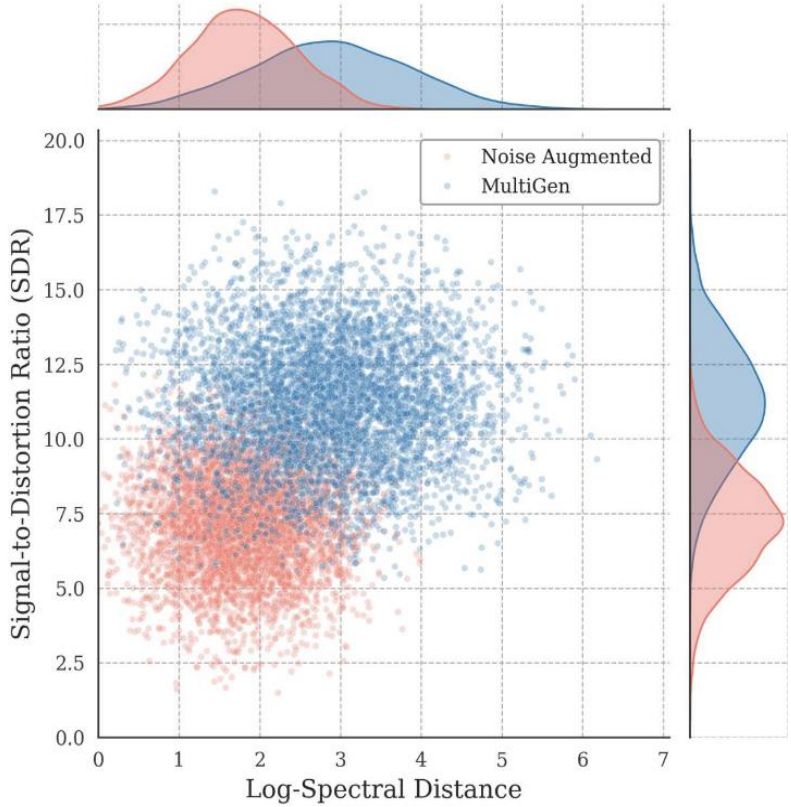
Metric

- **Diversity:** log-spectral distance (LSD) to real demos
- **Fidelity:** Signal-to-distortion ratio (SDR)
- **Usefulness:** policy performance, (2 opaque + 2 transparent)



Source: MultiGen, CoRL 2025, §4.4

MultiGen audio is more diverse and more accurate



Wide on x-axis (LSD): diverse

- Far from the original 10 demos: new sounds, not noisy copies

High on y-axis (SDR): accurate

- High SDR: little distortion vs. real sound

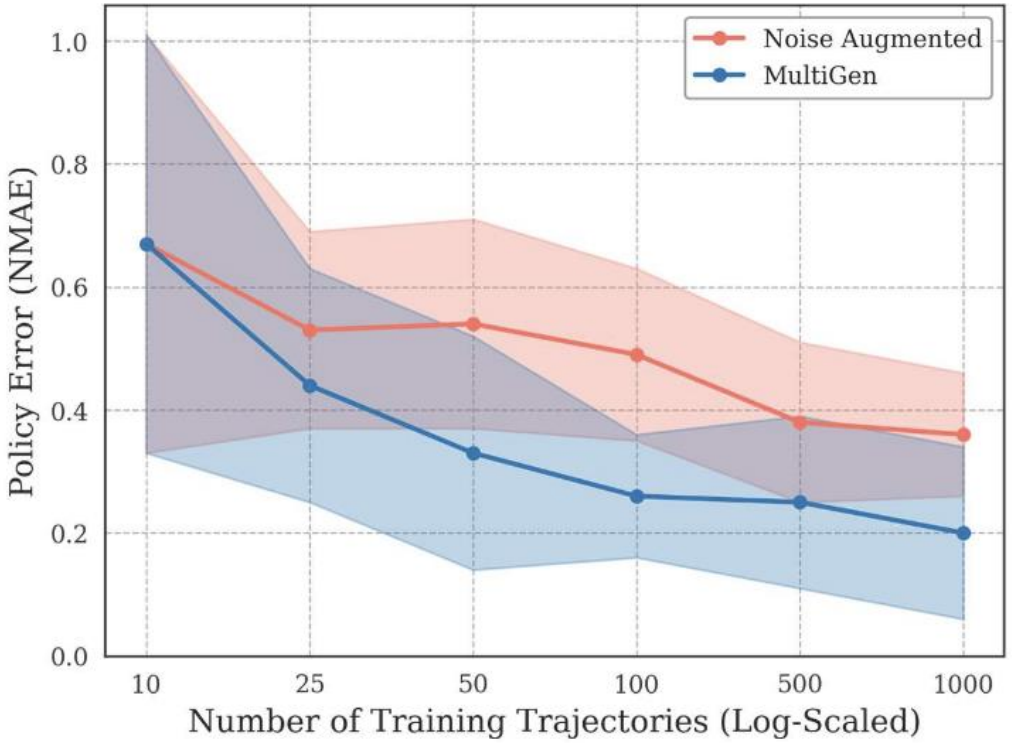
Upper right: diverse and accurate

- Far-from-demo samples + still keep high SDR

MultiGen audio is both more diverse and more accurate than augmentation!

Source: MultiGen, CoRL 2025, §4.4, Fig. 4a

MultiGen policies beat augmentation and scale better



Same policy trained w/ two different datasets

- Augmentation vs. MultiGen audio

MultiGen: lower error

- Better on 2 opaque + 2 transparent settings

MultiGen: keeps improving with more data

- NMAE falls further with more data
  (0.67 → 0.20 vs. 0.67 → 0.36)

MultiGen audio trains better policies, and the advantage grows with more data!

Source: MultiGen, CoRL 2025, §4.4, Fig. 4b

Finetuning MMAudio beats zero-shot on every metric

	Zero-shot	Finetuned	Change
Fidelity (SDR, higher is better)	7.8 dB	**11.3 dB**	+45%
Spectral error (LSD, lower is better)	1.95	**1.43**	−26.7%
Diversity (min-LSD, higher is better)	0.67	**1.02**	+52.2%
Policy error (NMAE, lower is better)	0.50	**0.37**	−26.0%

Setup: 100 simulation trajectories; zero-shot sees RGB only, finetuned sees RGB + segmentation masks

Finetuning adapts the pretrained model to pouring sounds, and the policy benefits

Source: MultiGen, CoRL 2025, Appendix B.2

Design choices

Choice	Why?
Video-conditioned V2A (MMAudio)	Language can’t control pouring’s level, volume
Finetune on 1,031 pouring videos	Pouring is long-tail in pretraining
+ SAMv2 segmentation (gated)	Bridging sim-vs-real video gap
Virtual camera on receiving container	Policy camera may be occluded; failed to generate
Diffusion Policy from scratch	Isolate the audio effect

Source: MultiGen, CoRL 2025, arXiv 2507.02864

Conclusion

Idea
A generative model adds audio, a sense that physics simulators cannot produce

Result
Policies w/ audio trained only on simulated data transfer zero-shot to real pouring, with no real robot data

Evidence
Audio lowers pouring error in all 8 conditions; generated audio beats noise augmentation

Scope
Shown for pouring; generative models can add new sensory modalities to simulation

Source: MultiGen, CoRL 2025, §5

Limitations

Real-to-sim setup

Needs approximate camera placement and robot match, so some manual setup

Container shape

Extreme or irregular shapes add unmodeled dynamics such as sloshing

Collisions

No collision avoidance in cluttered scenes

Source: MultiGen, CoRL 2025, §6

Our observations and open questions

Gaps we found

- **Gains within noise**
  Std. devs. overlap in some conditions
- **Confounded ablation**
  V2A finetuning and masks change together
- **Unclear what helps**
  Audio diversity vs. fidelity not separated

Open questions

- **Out of distribution**
  Do new containers and liquids work?
- **Masks ablation**
  Do segmentation masks bridge sim and real?
- **Other tasks**
  Does it help where sound is subtler?
- **Noisy environments**
  Does it work with robot and background noise?

Source: our reading of MultiGen, CoRL 2025, Table 1, Fig. 4, Appendix A–B

Towards the Vision-Sound-Language-Action Paradigm: The HEAR Framework for Sound-Centric Manipulation

Chang Nie et al. · Shanghai Jiao Tong Univ., Univ. of Cambridge

IJRR 2026 (accepted) (Q1 journal in Robotics)

arXiv 2603.16086 (Mar 2026)

TL;DR

Keep a streaming audio memory → act on sounds heard WHILE acting

Source: HEAR, IJRR 2026

Why short sounds get missed

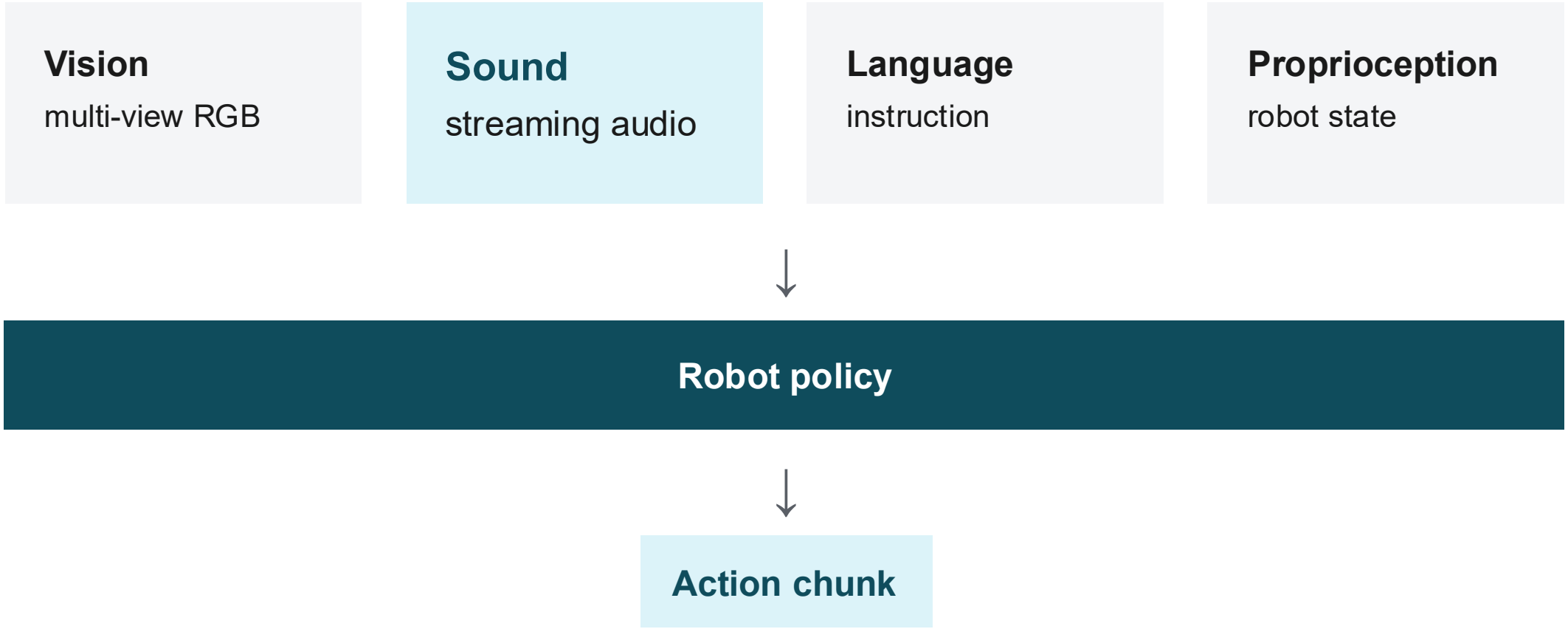


Open-loop action chunk + latency → no new observation during execution

Speech/ASR, waveform images, fixed pre-execution windows all share this blind spot

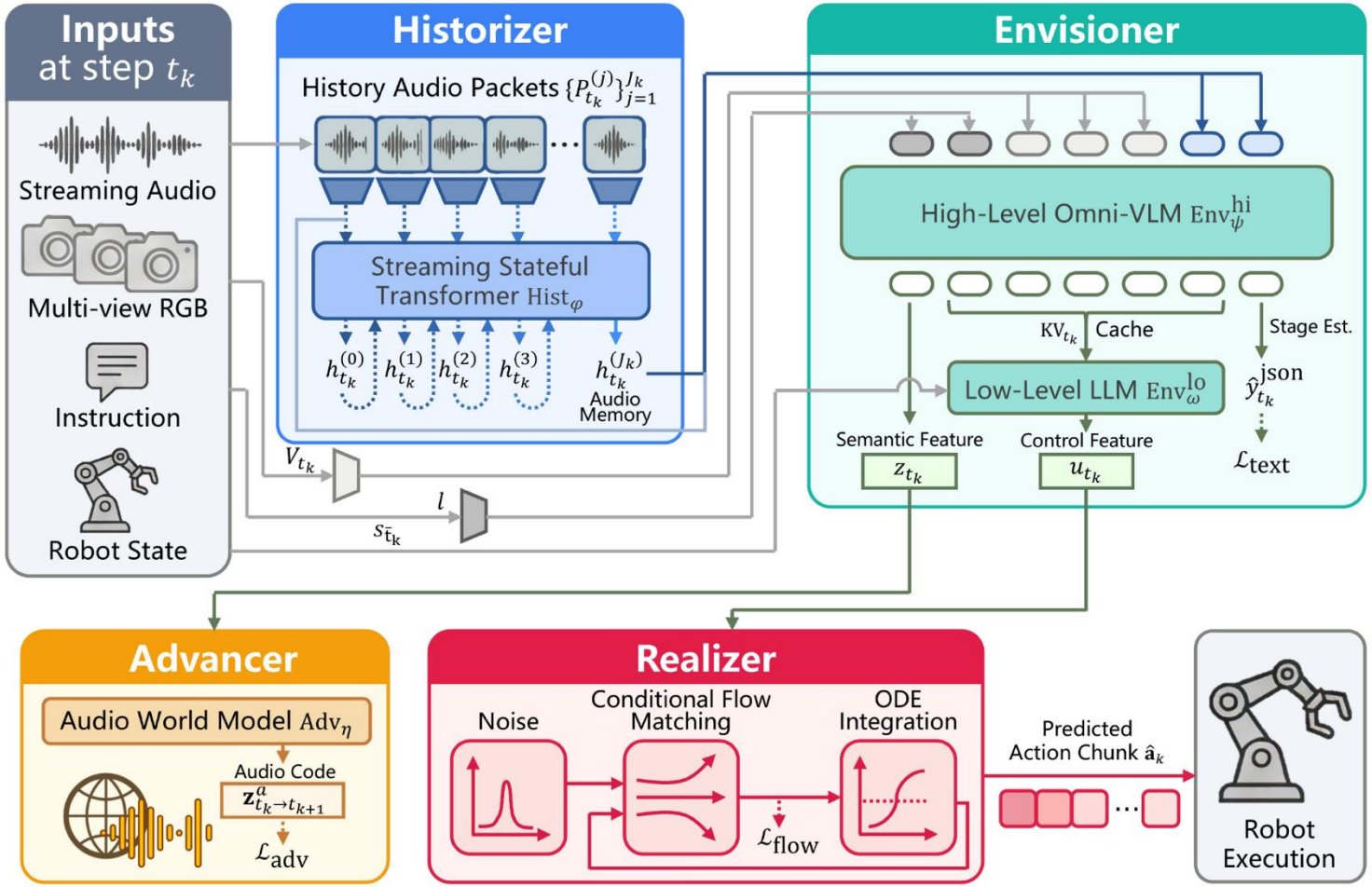
Source: HEAR, IJRR 2026, §1

From VLA to VSLA: vision-sound-language-action model



Source: HEAR, IJRR 2026, §1–3

HEAR: four modules in VSLA



Memory → Reasoning → (future prediction) → Action

Historizer
Streaming audio memory

Envisioner
Multimodal reasoning

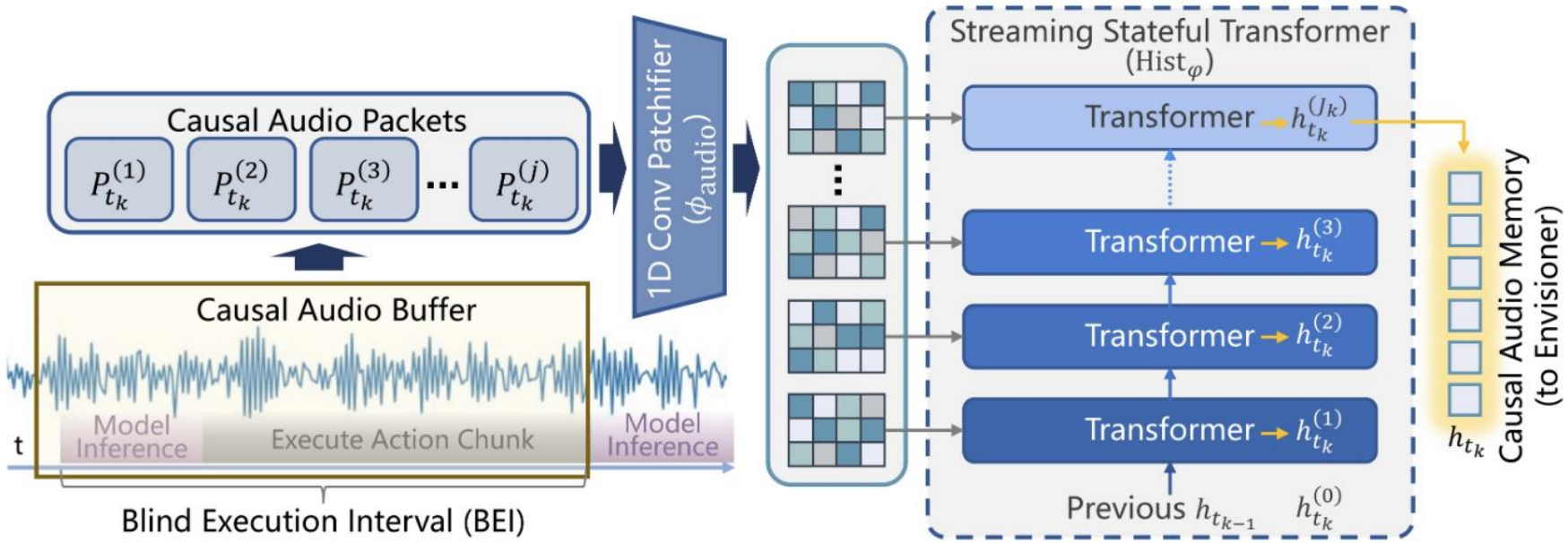
Advancer
Audio world model

Realizer
Flow-matching based action chunking

Source: HEAR, IJRR 2026, §3

Historizer: compact causal audio memory

A stateful transformer carries memory h forward, updating it with each new audio packet



$$h_{t_k}^{(j)} = \mathrm{Hist}_{\varphi}\left(h_{t_k}^{(j-1)},\ \phi_{\mathrm{audio}}\left(P_{t_k}^{(j)}\right)\right)$$

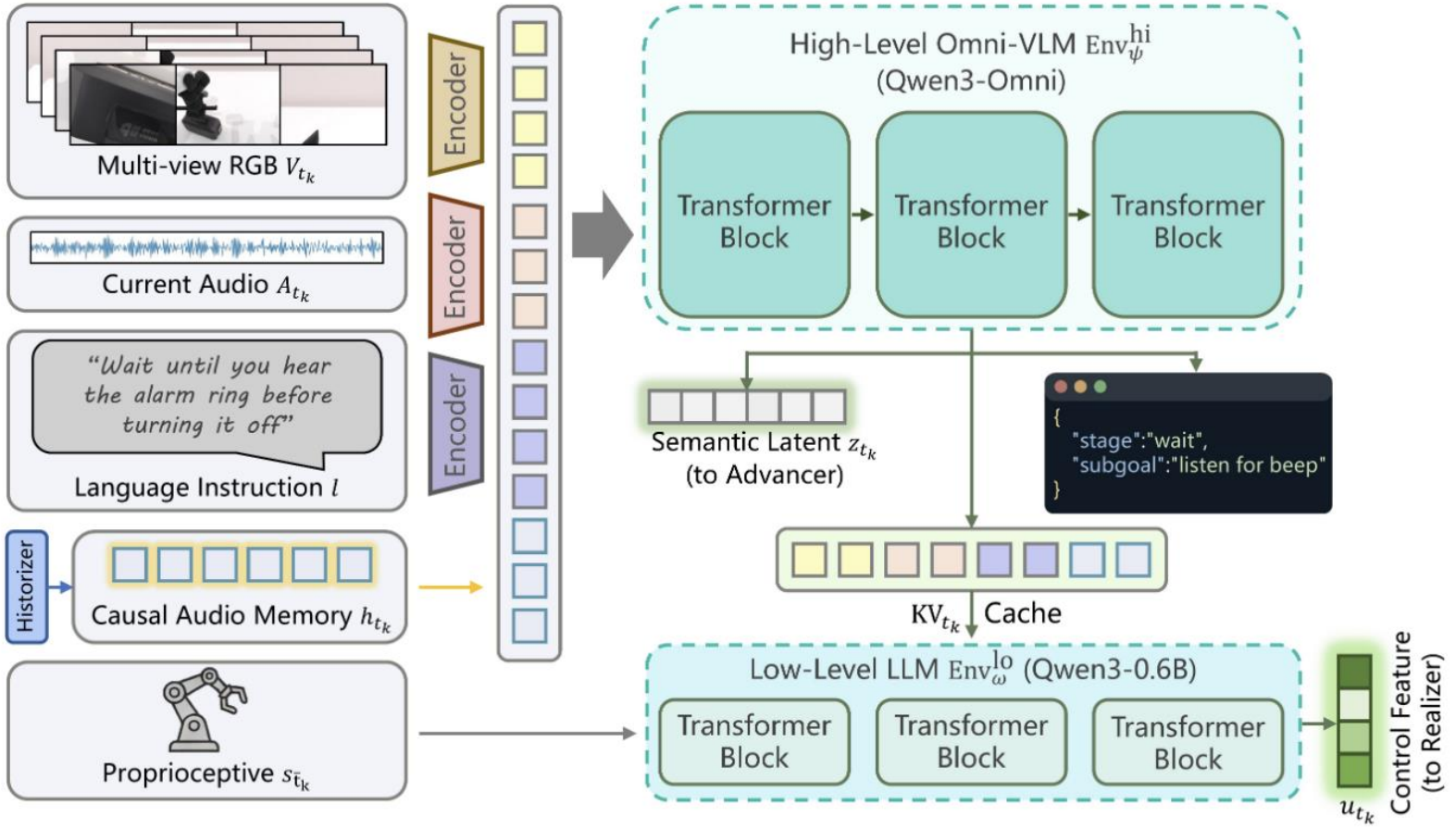
Input Previous memory h(j−1), audio packet P(j)

Output Updated memory h(j) → Envisioner

Source: HEAR, IJRR 2026, §4.2, Fig. 3

Envisioner: hierarchical reasoning

A large VLM reasons; a small LLM turns it into control



High level Qwen3-Omni

- z: semantic summary, sent to Advancer
- KV cache: context handed to low level
- Stage text: current subgoal

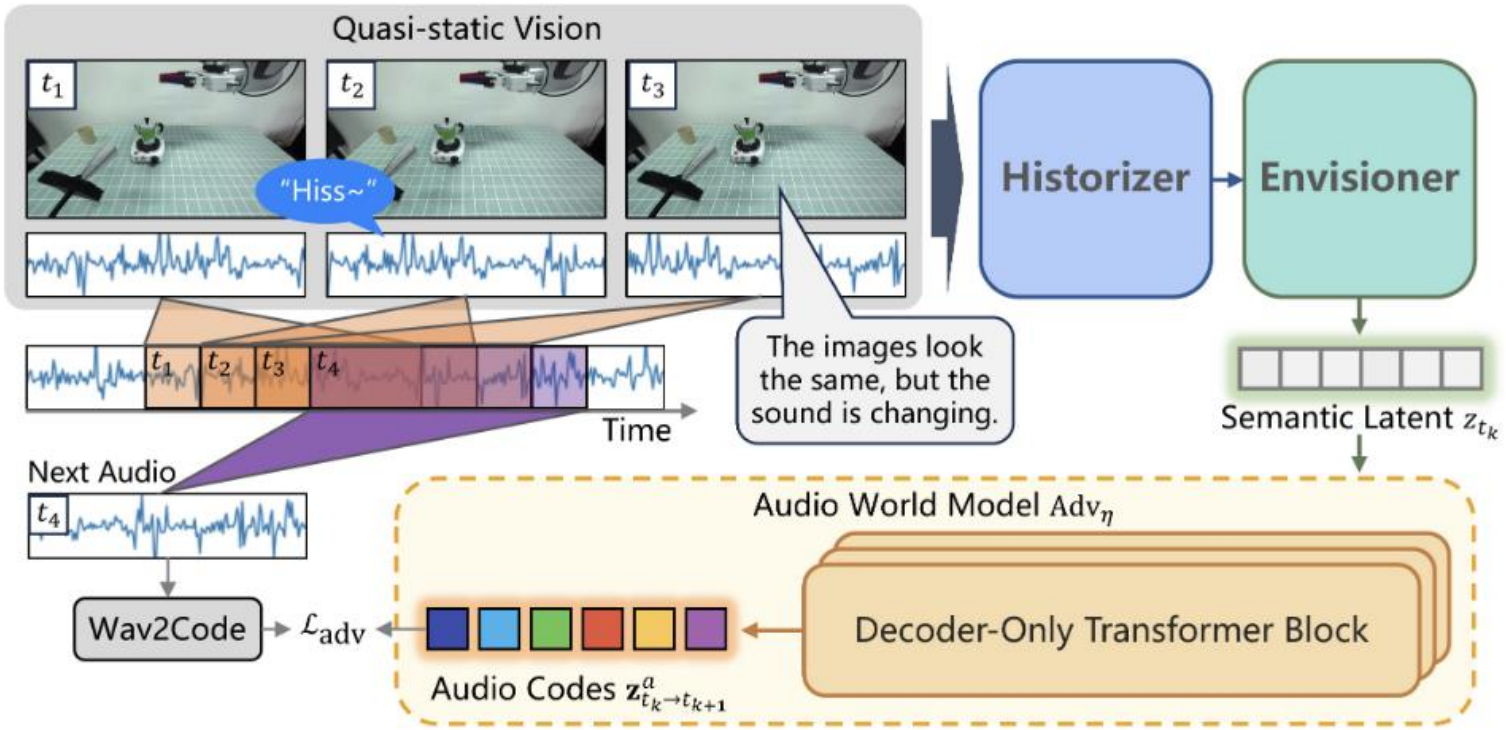
Low level Qwen3-0.6B

- u: control feature that conditions the Realizer

Source: HEAR, IJRR 2026, §3.2

Advancer: audio world model

An auxiliary audio-prediction task during training shapes the semantic latent z



Role

Predict the next audio codes from z (cross-entropy)

Why

Vision looks the same while waiting; sound keeps changing → z learns elapsed time

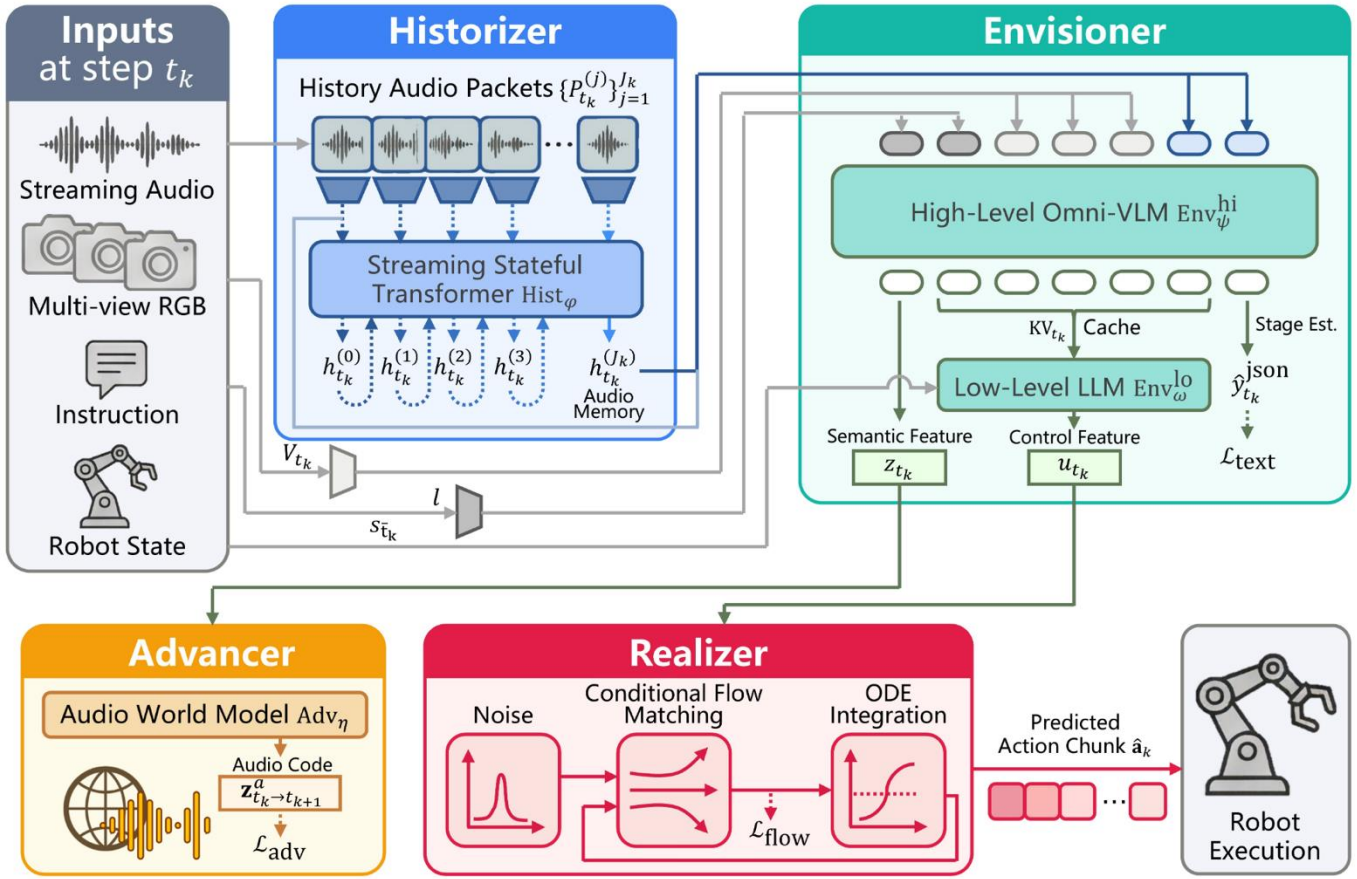
Cost

Training only, not deployed · 0.81 → 0.73 without it

Source: HEAR, IJRR 2026, §3.3, Fig. 5

Realizer: flow-matching policy

Flow matching turns the control feature u into smooth action chunks



Role

Generates the robot's action chunk

Input

- Semantic feature from High-level model
- Control feature from low-level model

Output

Smooth chunks of H = 16 steps

Source: HEAR, IJRR 2026, §3.4

OpenX-Sound: w/ audio dataset for pretraining

Add sound to existing robot data using a generative model

Open X-Embodiment

RGB, language, proprioception, actions dataset

→

+ V2A generated audio

video-to-audio models

↓

Human + automatic checks

fix or drop mis-timed audio

→

OpenX-Sound

multisensory pretraining

Source: HEAR, IJRR 2026, §4

Two-stage imitation learning

Pretrain on OpenX-Sound, then fine-tune per task

Training stage

Pretrained omni model

Qwen-Omni backbone

→

Stage 1 · Pretraining

OpenX-Sound (synthetic audio)

→

Stage 2 · Fine-tuning

Per task: HEAR-Bench or real-robot demos

Method

Offline imitation learning

Action chunk H = 16 (joint-position targets)

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{flow}} + w_{\text{adv}}\,\mathcal{L}_{\text{adv}} + w_{\text{text}}\,\mathcal{L}_{\text{text}},$$

Source: HEAR, IJRR 2026, §4.6, §6.1.2

HEAR-Bench: sound-causal tasks

7 simulated tasks (RoboTwin2), 4 cue types

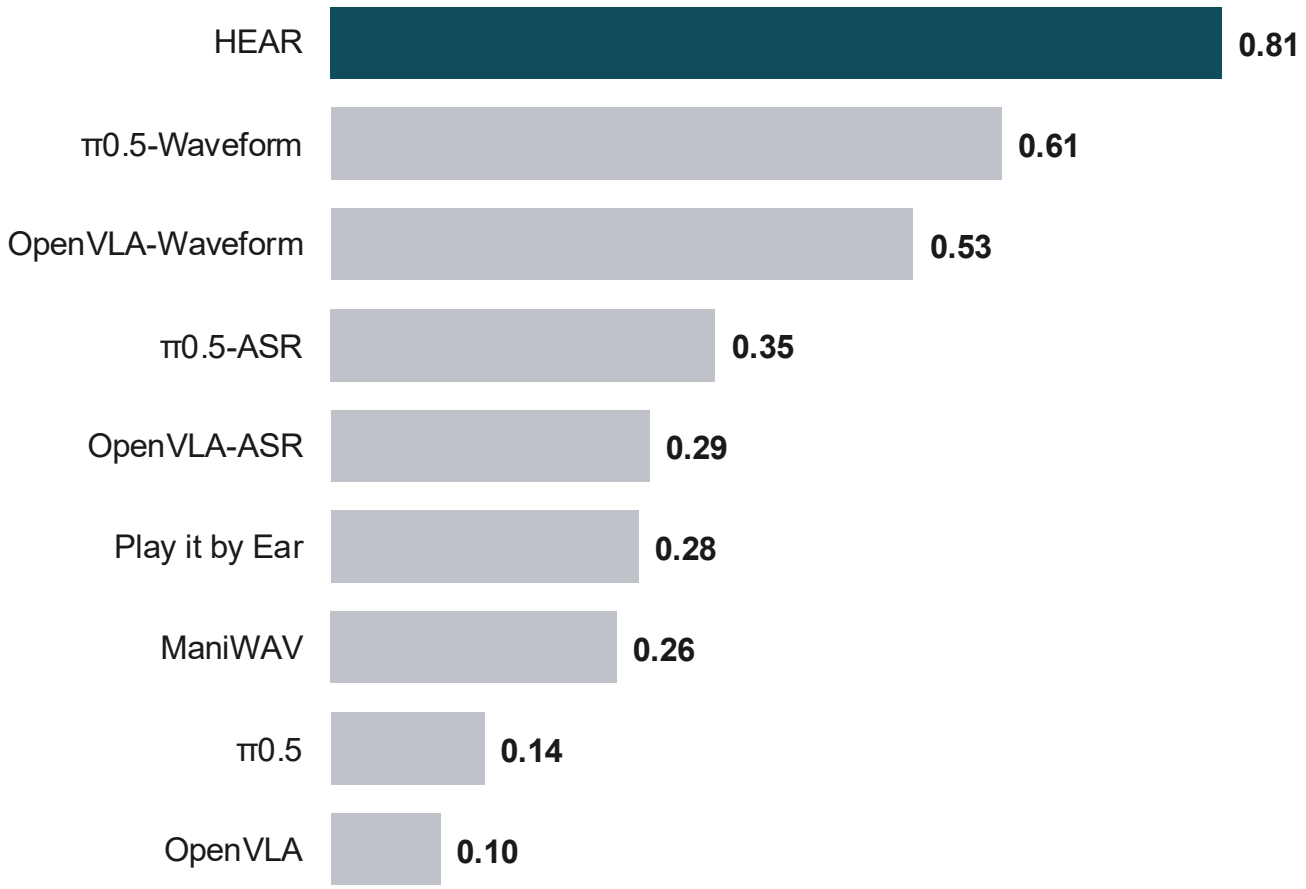
Cue type	Task	Robot must…	Cue
Event-triggered	**Alarm Clock**	Wait, press stop only after ringing starts	Ring onset
	Microwave	Wait, open the door after the beep	Brief “ding”
Speech	**Check Yes**	Keep or switch the bottle	Intonation
	Interrupt	Stop and return to home pose	Spoken “reset”
Continuous process	**Pour Water**	Stop at the target level (opaque cup)	Pouring sound evolving
	Boil Water	Intervene once it boils	Silence → simmer → boil
Physical interaction	**Check Materials**	Drop a cube, judge the plate, move it if wrong	Impact timbre

Success only if the goal is reached after the task’s sound cue has occurred

Simulated audio = task event clips mixed into a background stream, not physical acoustics

Source: HEAR, IJRR 2026, Table 1, §5.2, §6.2

HEAR-Bench: average success rate

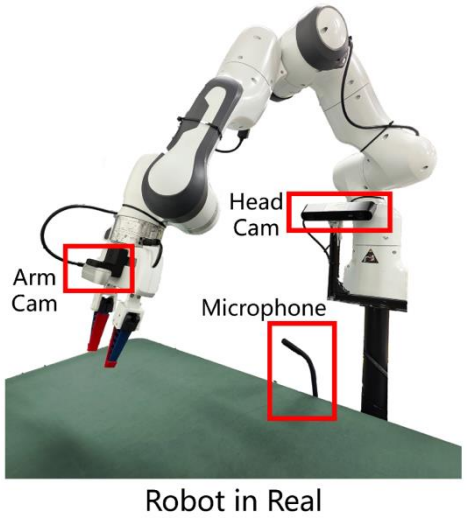


HEAR per task

Alarm Clock 0.91
Check Yes 0.89
Interrupt 0.88
Microwave 0.85
Check Materials 0.83
Boil Water 0.81
Pour Water 0.51

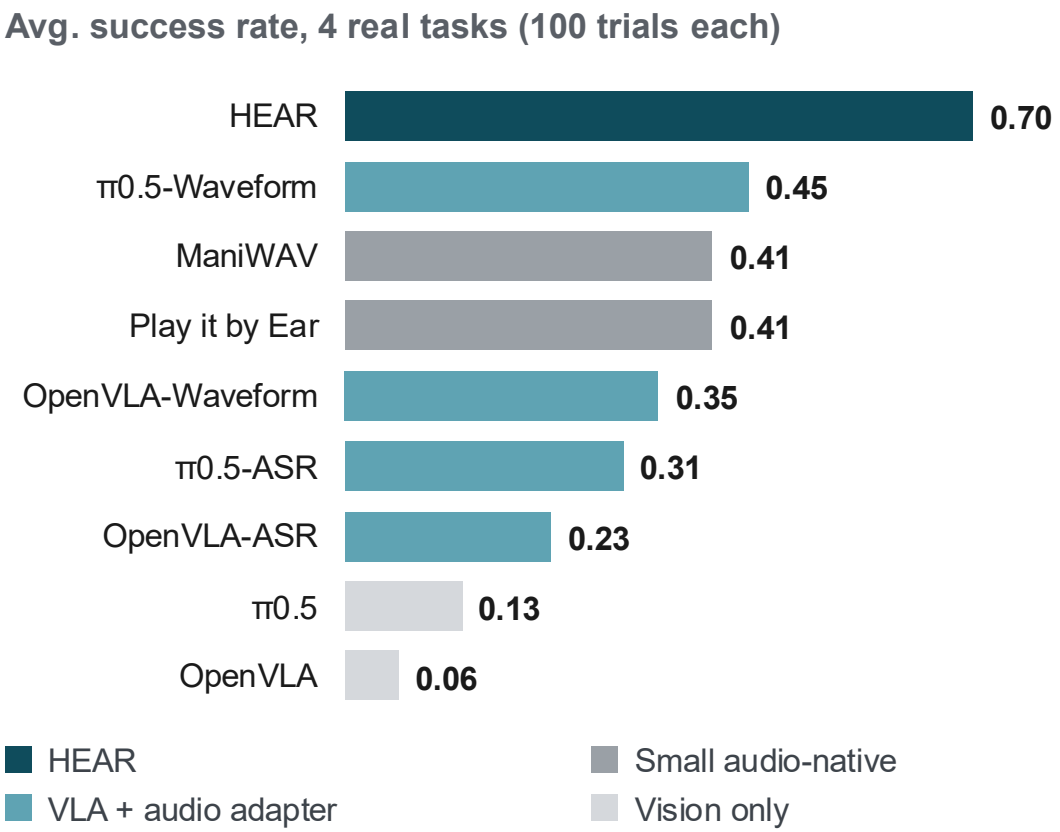
Source: HEAR, IJRR 2026, §6 (higher is better)

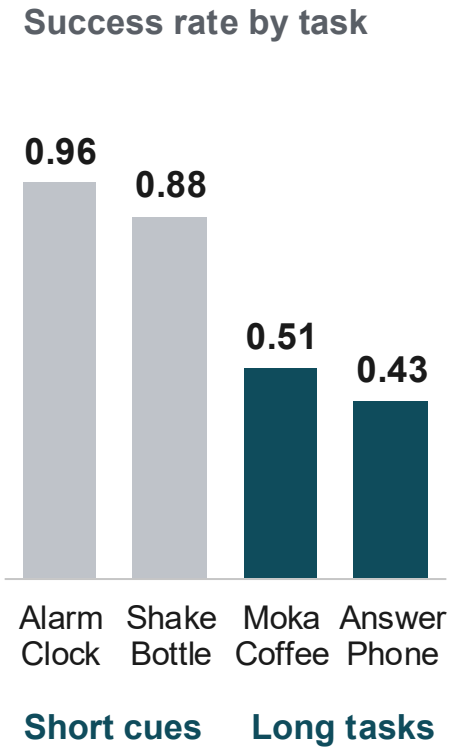
Real Franka: HEAR leads, but long tasks stay hard



Robot in Real

Franka Panda: head cam + arm cam + one workspace microphone



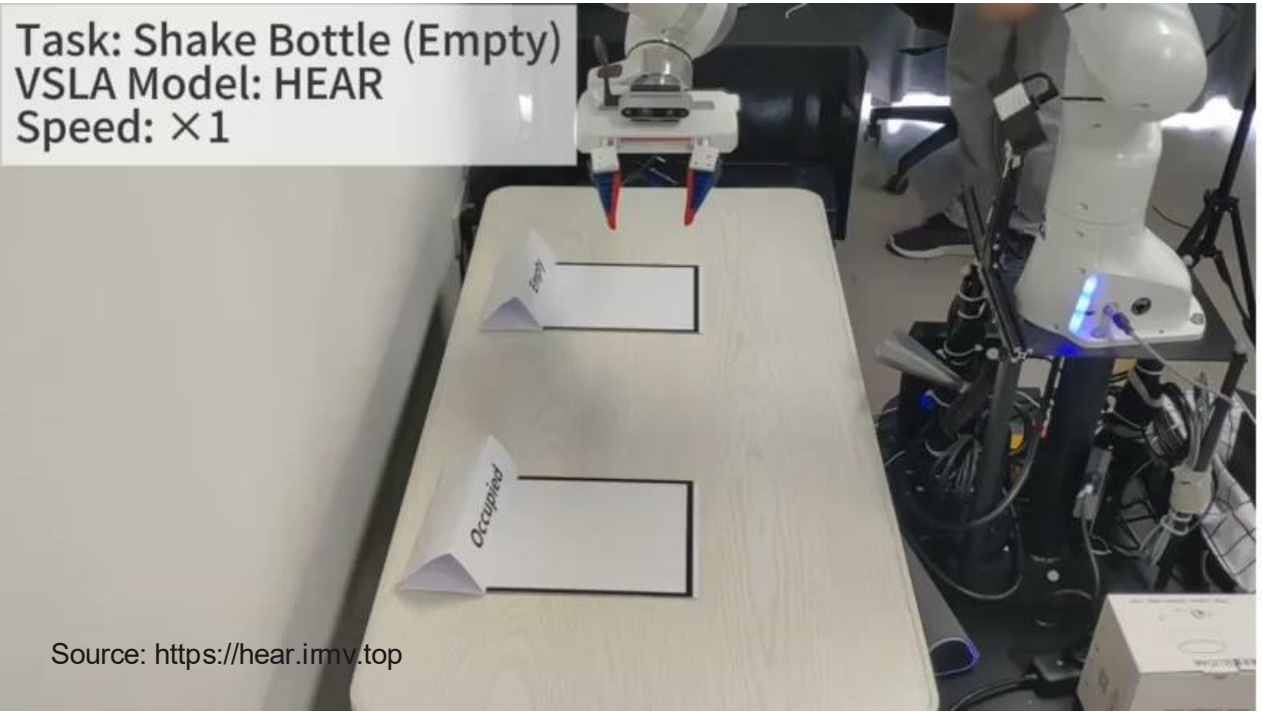


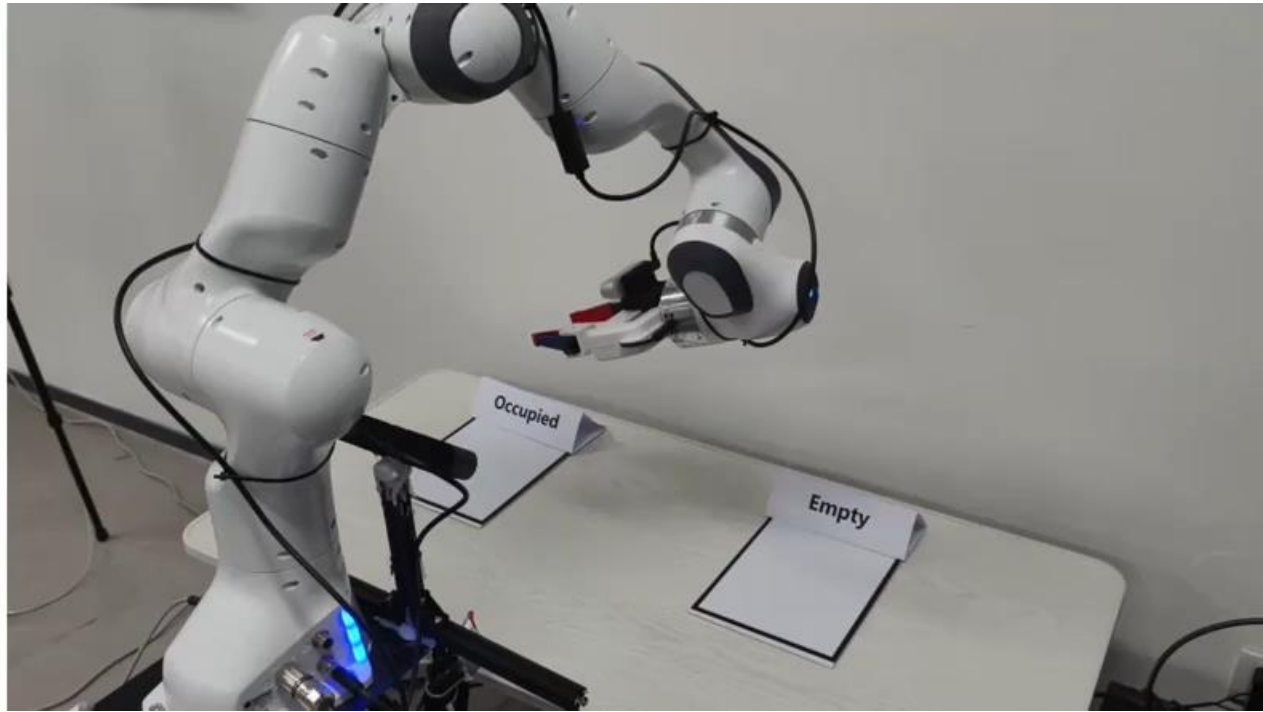
Calibration on the two long tasks: mic closer + gain (Moka 0.18 → 0.51), more dialogue demos (Phone 0.15 → 0.43)

Source: HEAR, IJRR 2026, Table 3, §6.3 · 100 trials per task







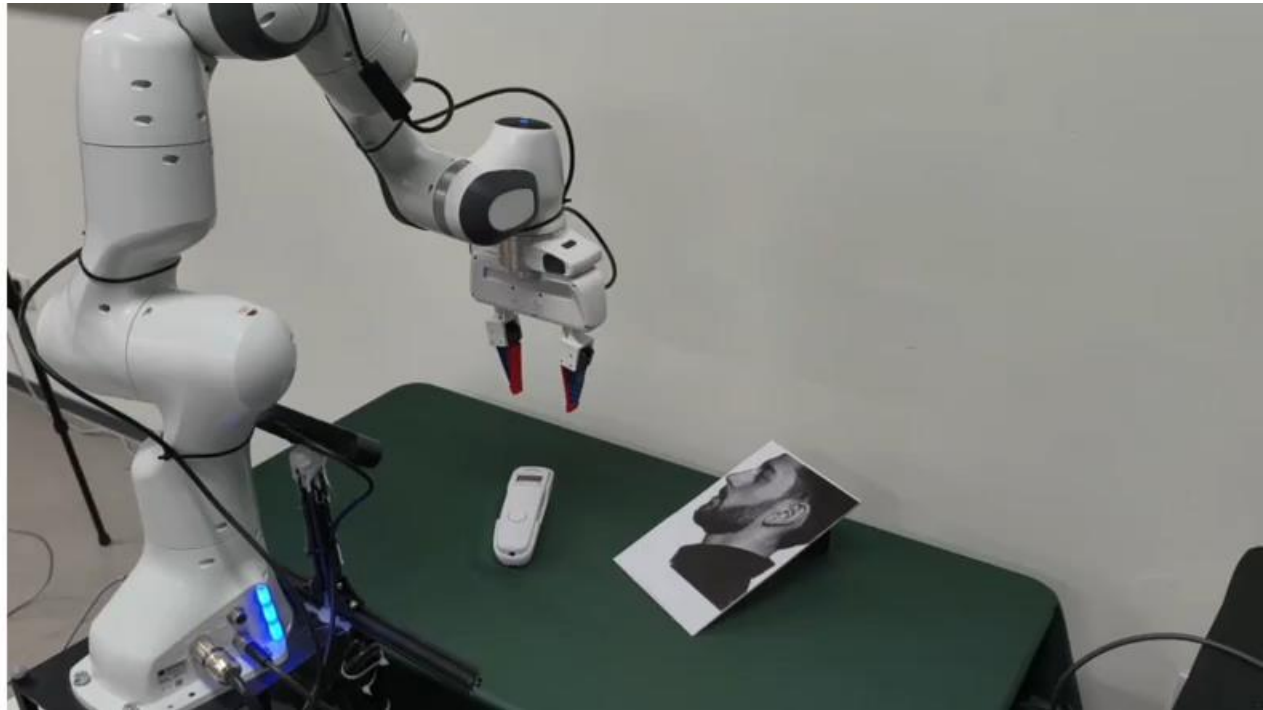


Source: https://hear.imv.top









Source: https://hear.imv.top

Conclusion

Idea
A causal audio memory keeps brief sounds available across open-loop action chunks

Result
0.81 on HEAR-Bench (simulation); 0.70 on a real Franka (calibrated comparison; 0.54 under a uniform setup)

Evidence
Beats waveform, ASR and compact audio-native baselines; removing the Historizer costs the most (0.81 → 0.57)

Scope
Seven simulated and four real tasks, one microphone; audio input alone is not enough: it needs causal persistence and temporal grounding

Source: HEAR, IJRR 2026, §6, §8

Limitations

Sensing
One workspace microphone; no source localization or separation of overlapping sounds

Training
Offline imitation only; robustness depends on demo diversity (ego-noise, reverberation)

Synthetic audio
Pretraining audio lacks room acoustics, mic placement, ego-noise and contact attack; real-robot fine-tuning is still needed

Long tasks
Moka Coffee and Answer Phone remain deployment-sensitive

Source: HEAR, IJRR 2026, §7.3–7.5

Our observations and open questions

Gaps we found

- **Ego-noise not measured**
  Only discussed qualitatively in the main text
- **Simulation can’t test it**
  HEAR-Bench audio is clip playback, no robot noise
- **Confounded ablation**
  GRU/EMA memories differ in capacity, so attention vs. size is not separated

Open questions

- **Ego-noise**
  Does the policy use its own motor sound as a cue?
- **Synthetic audio**
  How does generated-audio quality affect the policy?
- **Advancer**
  Time grounding, or just a richer latent?
- **Long tasks**
  Can causal memory scale to multi-stage tasks?

Source: our reading of HEAR, IJRR 2026, Tables 2–4, §6–7 (main text; supplement not read)

MultiGen vs. HEAR

	MultiGen	HEAR
Bottleneck	Sound is hard to simulate	Short sounds missed while acting
Contribution	Generative audio in simulation	Streaming audio memory
Audio source	MMAudio (finetuned on pouring)	V2A on Open X-Embodiment
Policy	Diffusion Policy (from scratch)	Omni LLM + flow matching
Evaluation	Real pouring; NMAE	HEAR-Bench + Franka; success rate

Source: MultiGen, CoRL 2025; HEAR, IJRR 2026

Common thread: generated audio

MultiGen

Generated audio = training data

Solves the data side

HEAR

Generated audio = pretraining data

Solves the model side

Complementary: streaming memory model trained on generated audio

Source: MultiGen, CoRL 2025; HEAR, IJRR 2026

Summary

MultiGen

Data

Generate audio in simulation → zero-shot real transfer

HEAR

Architecture

Streaming audio memory → act on sounds while acting

Audio-aware robot policies need both: realistic sound data and persistent auditory memory

Source: MultiGen, CoRL 2025; HEAR, IJRR 2026

Open questions

1. Can generative models produce physically accurate and realistic sound?
2. How robust is audio-based manipulation to robot ego-noise and background noise?
3. What is the best way to feed sound to a robot policy?
4. Are memory and a multi-module architecture effective?

Source: MultiGen, CoRL 2025; HEAR, IJRR 2026

References

[1] R. Wang et al. The Sound of Simulation: Learning Multimodal Sim-to-Real Robot Policies with Generative Audio. CoRL 2025. arXiv:2507.02864.

[2] C. Nie, T. Deng, G. Wang, Z. Liu, H. Wang. Towards the Vision-Sound-Language-Action Paradigm: The HEAR Framework for Sound-Centric Manipulation. IJRR 2026. arXiv:2603.16086.

[3] Project page: The Sound of Simulation (MultiGen). https://multigen-audio.github.io

[4] Project page: HEAR. https://hear.irmv.top

THANK YOU

Questions?